Trigonometry Selective

Part A: Lengths and Angles

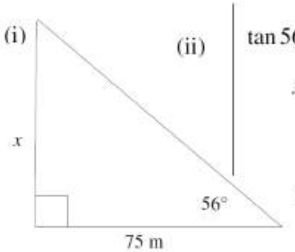
Q 1 ACE

A building on level ground, casts a shadow of length 75 metres when the angle of elevation of the sun is 56°.

(i) Draw a diagram to show this information. **1**

(ii) Calculate the height of the building. Answer correct to the nearest one-tenth of a metre. **2**

(i)



(ii)

$$\tan 56 = \frac{x}{75}$$

$$x = 75\tan 56$$

$$= 111.1920726...$$

$$\approx 111.2 \text{ m}$$

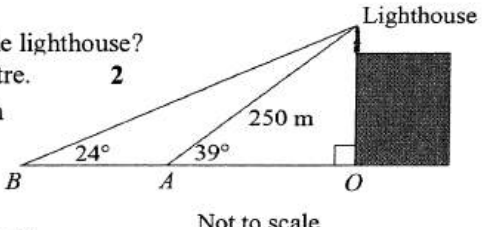
Height of the building is 111.2 metres

Q 2 ACE

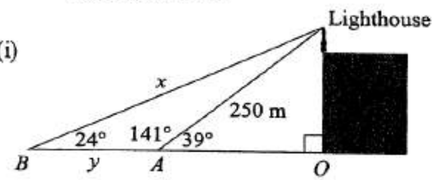
Two ships are both due east of a lighthouse. The angle of elevation from Ship A is 39° and from Ship B is 24°. Ship A is exactly 250 metres from the lighthouse.

(i) How far is ship B from the lighthouse? Answer to the nearest metre. **2**

(ii) Find the distance between the two ships. Answer to the nearest metre. **2**



(i)



1

$$\frac{x}{\sin 141^\circ} = \frac{250}{\sin 24^\circ}$$

$$x = \frac{250 \times \sin 141^\circ}{\sin 24^\circ}$$

$$= 386.8107298...$$

$$\approx 387 \text{ m}$$

Ship B is 387 metres from the lighthouse.

1(f) (ii)

$$y^2 = 387^2 + 250^2 - 2\times 387 \times 250 \times \cos 15^\circ$$

$$= 25\,307.3038...$$

$$y = 159.0826949...$$

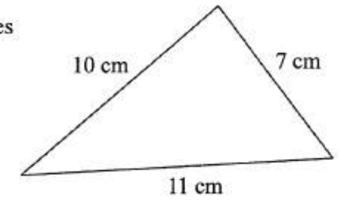
$$\approx 159$$

Ships are 159 metres apart.

Q 3 ACE

The following triangle has sides 7 cm, 10 cm and 11 cm.

Not to scale



Angle A is the smallest angle. Which of the following expressions is correct for angle A?

(A) $\cos A = \dfrac{7^2 + 11^2 - 10^2}{2 \times 7 \times 11}$

(B) $\cos A = \dfrac{10^2 + 7^2 - 11^2}{2 \times 7 \times 10}$

(C) $\boxed{\cos A = \dfrac{10^2 + 11^2 - 7^2}{2 \times 10 \times 11}}$

(D) $\cos A = \dfrac{10^2 + 7^2 - 11^2}{2 \times 10 \times 11}$

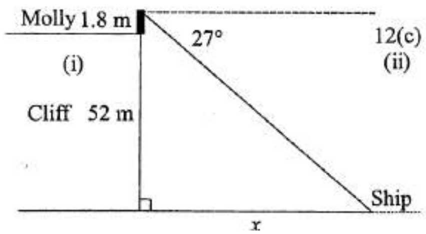
Q 4 ACE

Molly is standing at the top of a vertical cliff. The cliff is 52 metres above sea level and she is 1.8 metres tall. Molly observes a ship out to sea with an angle of depression of 27°.

(i) Draw a neat sketch showing this information. **1**

(ii) How far is the ship from the base of the cliff? **2**

(i)



12(c) (ii)

Complement of 27° is 63°

$$\tan 63^\circ = \frac{x}{52 + 1.8}$$

$$x = 53.8\tan 63^\circ$$

$$= 105.5884452... \approx 106 \text{ m}$$

Ship is 106 metres from the base of the cliff.

Q 5

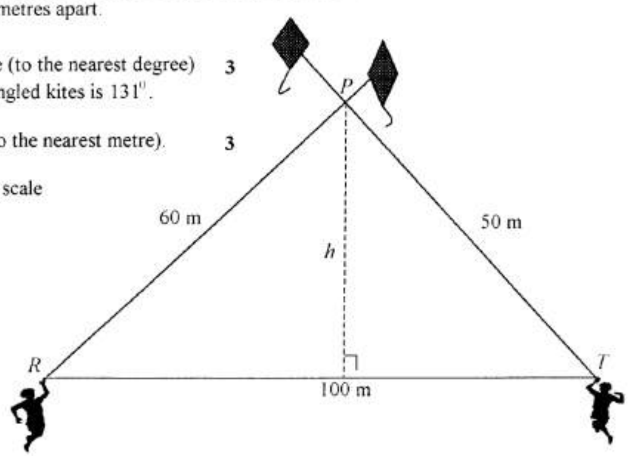
AMP

Ramish and Tran have their kites tangled at point P in a kite flying exhibition.
Point P is 60 metres from Ramish at R and 50 metres from Tran at T.
Ramish and Tran are 100 metres apart.

(i) Show that the angle (to the nearest degree) between the two tangled kites is 131°. **3**

(ii) Find the height h (to the nearest metre). **3**

Diagram not to scale



(a) (i) Using Cosine rule in ΔPTR

$100^2 = 50^2 + 60^2 - 2\times 50\times 60\times \cos\angle P$ ✓

$\therefore\ \cos\angle P = \dfrac{2500+3600-10000}{6000}$ ✓

$= -0.65$

$\therefore\ \angle P = \cos^{-1}(-0.65) \approx 131^{\circ}$ as required. ✓

(ii) In ΔPTR, using the sine rule

$\dfrac{\sin\angle T}{60} = \dfrac{\sin\angle P}{100}$ ✓

$\sin\angle T = \dfrac{60\times\sin\angle P}{100}$

$\therefore\ \angle P = 27^{\circ}8'$ ✓

In ΔPST, using sine ratio

$\sin 27^{\circ}8' = \dfrac{h}{50}$

$\therefore\ h = 50\sin 27^{\circ}8'$

$= 23$ m (to the nearest metre). ✓

Q6

ASCHAM

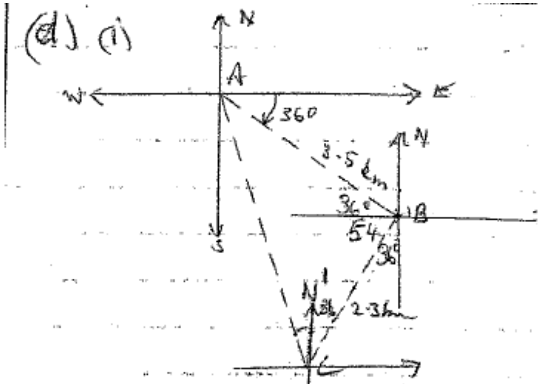
Mary walks from point A to point B for 3.5 km on a bearing of 126°. She then changes direction and walks to point C, 2.3 km away on a bearing of 216°, and stops.

i) Draw a diagram showing all details of her journey. **1**

ii) Find the distance AC. Give your answer correct to one decimal place. **2**

iii) What is the bearing of point A from point C? Give your answer to the nearest minute. **2**

(d) (i)



(ii)

$\angle ABC = 90^{\circ}$

$\therefore\ AC^2 = 3.5^2 + 2.3^2$

$AC = 4.2$ km. (1 d.p.)

Q 7
ASCHAM

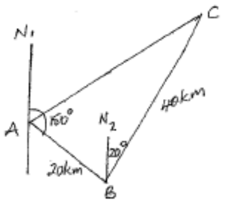
Two geologists on a large flat mining claim drive 20 km from point A on a bearing of 150^0 T to point B. They then drive 40 km on a bearing of 020^0 T to point C.

i) Copy the above diagram into your examination booklet and fill in the data (2)

ii) Show that $\angle ABC = 50^0$ (2)

iii) Find the distance of point C from point A to the nearest kilometre. (2)

b) i)



(2)

ii) $\angle N_2BA = 30°$ (co-int $\angle$s $N_1A \parallel N_2B$) ✓

$\therefore \angle ABC = 30° + 20°$

$= 50°$ ✓ (2)

iii) $AC^2 = 20^2 + 40^2 - 2 \times 20 \times 40 \cos 50$ ✓

$= 971 \cdot 5398 \ldots$

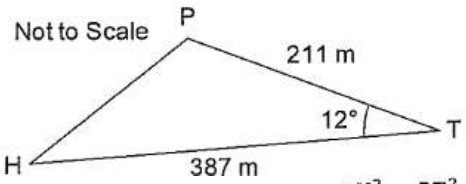
$AC = 31 \cdot 16 \ldots$ ✓

$\therefore$ distance of C from A is 31 km (to n km) (2)

Q 8
BAULKHAM

On a golf course, the distance from a tee T to the hole H is 387 metres. A golfer's ball comes to rest at P, 211 metres from T. Given that $\angle PTH = 12°$ how far is it from P to H. 3

Not to Scale



$PH^2 = PT^2 + HT^2 - 2.PT.HT.\cos\angle PTH$

$OR\ \ (q^2 = p^2 + h^2 - 2ph\cos T)$ 1

$PH^2 = 211^2 + 387^2 - 2 \times 211 \times 387 \cos 12°$

$PH^2 = 34544.80273$ 1

$PH = 185.862322\ldots$ 1

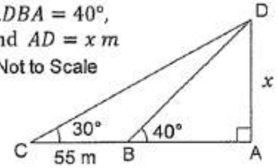
Q9
BAULKHAM

In the diagram $\angle DAC = 90°, \angle DBA = 40°$, $\angle DCA = 30°$, $CB = 55\ m$ and $AD = x\ m$

i) Find DB 2

ii) Hence find x 2

Not to Scale



i) $\angle CDB = 10°$

(ext $\angle$ = sum interior opp $\angle$'s) 1

$$\frac{DB}{\sin 30°} = \frac{CB}{\sin 10°}$$

$DB = 158.366188\ldots$ 1 158 OK

ii) $\sin 40° = \frac{x}{DB}$ 1

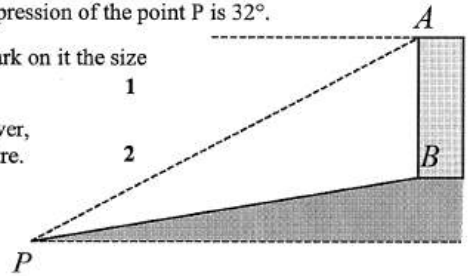
$x = 101.795823\ldots$ 1

Q10
INDEPENDENT

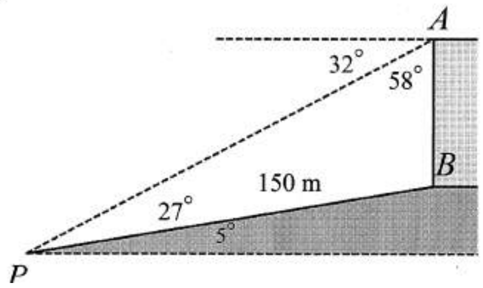
A point P is at the foot of the hill which is inclined at $5°$ to the horizontal. The base B, of a tower AB, is situated 150 metres up the incline of the hill from P. From the top of the tower the angle of depression of the point P is 32°.

(i) Copy the diagram and mark on it the size of $\angle APB$ and of $\angle PAB$. 1

(ii) Find the height of the tower, correct to the nearest metre. 2



1)



ii) Using the sin rule: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

$$\frac{h}{\sin 27} = \frac{150}{\sin 58}$$

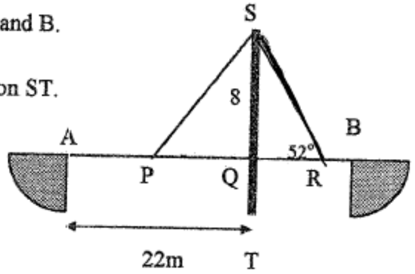
$$h = \frac{150 \sin 27}{\sin 58}$$

$h = 80\ m$ (nearest m)

Q11
ASCHAM

A horizontal bridge is built between points A and B. The bridge is supported by cables SP and SR, which are attached to the top of a vertical pylon ST.

The section of the pylon, SQ, above the bridge is 8 metres long and <SRQ = 52°.
The distance AQ is 22 metres and P is the midpoint of AQ.



i. Find the length of the cable SR to the nearest centimetre. 3 m

ii. Find the size of < SPQ to the nearest degree. 2 m

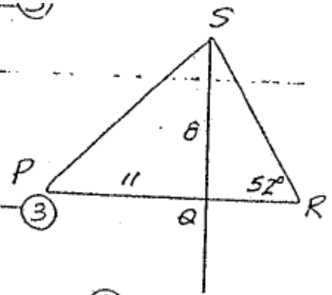
i. $\frac{8}{SR} = \sin 52°$

$SR = 10.152\ldots\ m$

$= 1015\ cm$ — 3

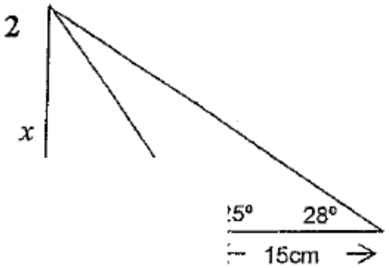
ii. $\tan S\hat{P}Q = 8/11$

$\therefore S\hat{P}Q = 36°$ — 2



Q12
BAULKHAM

Find x correct to 3 significant figures 2



Find AC first

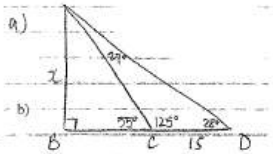
$\frac{AC}{\sin 28°} = \frac{15}{\sin 27°}$

$AC = 15.51149957$

$\sin 55° = \frac{x}{15.51149957}$

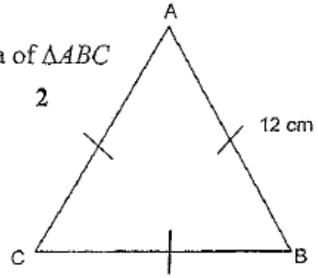
$x = 12.70627658$

$x = 12.7\ cm$ (3 sig fig)



Q13
BAULKHAM

Find the exact area of ΔABC 2



Each angle = 60°

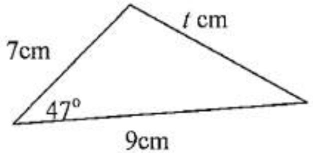
$A = \frac{1}{2} \times a \times b \times \sin C$

$= \frac{1}{2} \times 12 \times 12 \times \sin 60$

$= 72 \times \frac{\sqrt{3}}{2}$

$= 36\sqrt{3}\ cm^2$

Q14
GIRAWEEN



Find:

(i) the value of t, correct to one decimal place. 2

(ii) area of the triangle, correct to one decimal place. 2

i) $t^2 = 7^2 + 9^2 - 2(7)(9)\cos 47°$

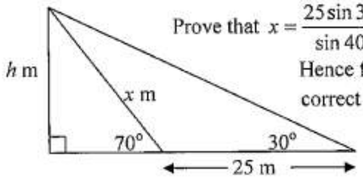
$t^2 = 44.068$

$\therefore t = 6.6\ cm$

ii) $A = \frac{1}{2}(7)(9)\sin 47°$

$= 23.0\ cm^2$

Q15
GIRAWEEN



Prove that $x = \dfrac{25\sin 30}{\sin 40}$.
Hence find the value of h, correct to one decimal place. 3

i) $\dfrac{x}{\sin 30^\circ} = \dfrac{25}{\sin 40^\circ}$

$\therefore x = \dfrac{25\sin 30^\circ}{\sin 40^\circ}$

$\sin 70^\circ = \dfrac{h}{\frac{25\sin 30^\circ}{\sin 40^\circ}}$

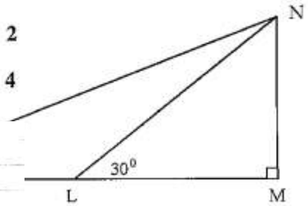
$h = \dfrac{25\sin 30^\circ}{\sin 40^\circ} \times \sin 70^\circ$

$h = 18 \cdot 3m$

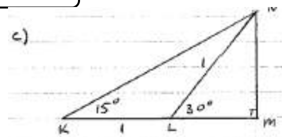
Q16
GIRAWEEN

(i) If $KL = 1$ metre, explain why LN also equals 1 metre. 2

(ii) Use the sketch to deduce that: $\tan 15^0 = 2 - \sqrt{3}$ 4



c)



i) $\angle KNL = 15^\circ$ (sum of interior opposite angles equals exterior angle of $\triangle KLN$)

$\therefore \triangle KLN$ is isosceles

$\therefore LN = 1$ metre

ii) in $\triangle LNM$

$\cos 30^\circ = \dfrac{LM}{1}$

$\dfrac{\sqrt{3}}{2} = LM$

$\sin 30^\circ = \dfrac{MN}{1}$

$\dfrac{1}{2} = MN$

in $\triangle KNM$

$\tan 15^\circ = \dfrac{MN}{KM}$

$= \dfrac{\frac{1}{2}}{1 + \frac{\sqrt{3}}{2}}$

$= \dfrac{1}{2+\sqrt{3}} \times \dfrac{2-\sqrt{3}}{2-\sqrt{3}}$

$= \dfrac{2-\sqrt{3}}{4-3}$

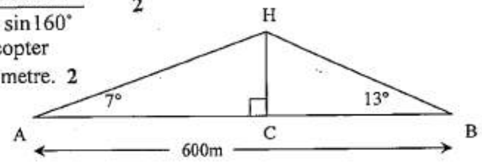
$\tan 15^\circ = 2-\sqrt{3}$

Q17
BAULKHAM

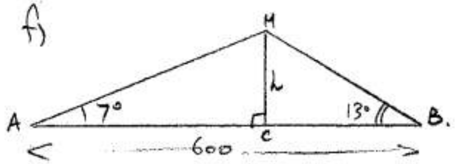
A helicopter is hovering above a straight, horizontal road AB of length 600 metres. The angles of elevation of H from A and B are 7° and 13° respectively. The point C lies on the road directly below H.

(i) Use the sine rule to show that $HB = \dfrac{600\sin 7^\circ}{\sin 160^\circ}$ 2

(ii) Hence find the height CH of the helicopter above the road, correct to the nearest metre. 2



f)



$\therefore \angle AHB = 180 - 13 - 7$

$= 160^\circ$

i) $\dfrac{HB}{\sin 7^\circ} = \dfrac{600}{\sin 160^\circ}$

$\therefore HB = \dfrac{600\sin 7^\circ}{\sin 160^\circ}$ (2)

ii) $\therefore \sin 13^\circ = \dfrac{h}{HB}$

$h = HB\sin 13^\circ$

$h = \dfrac{600\sin 7^\circ \sin 13^\circ}{\sin 160^\circ}$

$h = 48.09..$

$h \doteqdot 48m$ (to nearest metre) (2)

Q18

GOSFORD

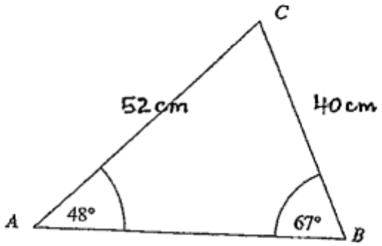
What is the correct expression for the area of triangle ABC?

A. $\frac{1}{2} \times 52 \times 40 \times \cos 67°$

B. $\frac{1}{2} \times 52 \times 40 \times \sin 67°$

C. $\frac{1}{2} \times 52 \times 40 \times \cos 65°$

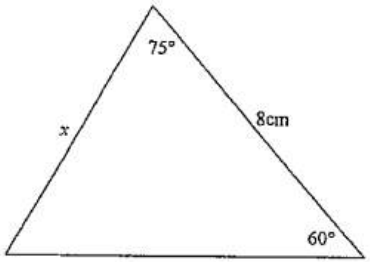
D. $\boxed{\frac{1}{2} \times 52 \times 40 \times \sin 65°}$



Q19

GIRAWEEN

Find the exact value of x in the triangle below.



NOT TO SCALE

Third angle in triangle is $45°$

$$\frac{x}{\sin 60°} = \frac{8}{\sin 45°}$$

$$x = \frac{8 \sin 60°}{\sin 45°}$$

$$= \frac{8 \times \frac{\sqrt{3}}{2}}{\frac{1}{\sqrt{2}}}$$

$$= 4\sqrt{3} \times \frac{\sqrt{2}}{1}$$

$$= 4\sqrt{6}$$

Q20

KINGS

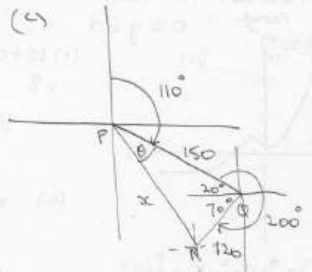
A ship leaves a port P and travels 150 nautical miles to port Q on a bearing of $110°$. It then travels 120 nautical miles to port R on a bearing of $200°$.

(i) Explain why $\angle PQR = 90°$. **2**

(ii) Find, correct to the nearest degree, the bearing of port R from port P. **2**

(iii) How far is port R from port P, to the nearest nautical mile? **1**

(c)



(i) $\angle PQR = 20°$ (alt $\angle$'s $\parallel$)

$+ 70$ $(270 - 200)$

$= 90°$

(ii) Bearing of R from P $= 110 + 38°39'$

$= 149°$

(iii) $x^2 = 120^2 + 150^2$

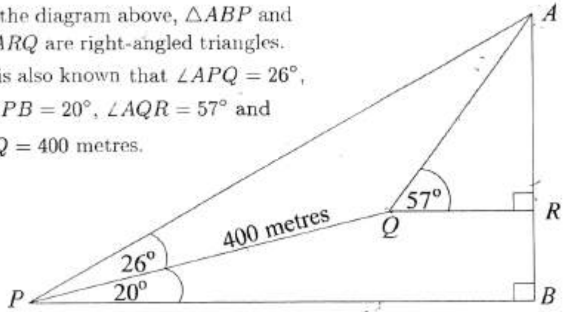
$= 192$ nm

Q21
GOSFORD

The bearing of Town A from Town B is 171°.
What is the bearing of Town B from Town A?

$\boxed{\text{(A)} \quad 351°}$		(C)	171°
(B)	009°	(D)	261°

Q22
KINGS



In the diagram above, $\triangle ABP$ and $\triangle ARQ$ are right-angled triangles. It is also known that $\angle APQ = 26°$, $\angle QPB = 20°$, $\angle AQR = 57°$ and $PQ = 400$ metres.

(i) Show that $\angle AQP = 143°$.

(ii) Find the length AB, correct to the nearest metre.

a) (i) $\angle PQR = 360 - 200$ (angle sum PQRB)
$= 160°$ ✓
$\therefore \angle AQP = 360 - (160 + 57)$ (angles at point)
$= 143°$ ✓

(ii) $\dfrac{AP}{\sin 143} = \dfrac{400}{\sin 11}$ ✓

$AP = \dfrac{400 \sin 143}{\sin 11}$

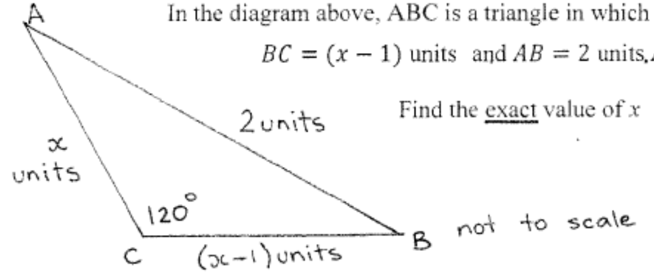
$= 1261.607\ldots$

$\sin 46 = \dfrac{AB}{AP}$ ✓

$\therefore AB = \dfrac{400 \sin 143}{\sin 11} \cdot \sin 46$

$= 907.52\ldots$ ✓ $= 908$ m

Q23
KINGS



In the diagram above, ABC is a triangle in which $AC = x$ units, $BC = (x-1)$ units and $AB = 2$ units, $ACB = 120°$

Find the <u>exact</u> value of x 3

$2^2 = x^2 + (x-1)^2 - 2x(x-1)\cos 120°$

$4 = x^2 + x^2 - 2x + 1 - 2x(x-1)(-\tfrac{1}{2})$

$4 = 2x^2 - 2x + 1 + x^2 - x$

$0 = 3x^2 - 3x - 3$

$0 = x^2 - x - 1$

$x = \dfrac{1 \pm \sqrt{1 - 4\times1\times-1}}{2}$

$x = \dfrac{1 \pm \sqrt{5}}{2}$ since $x > 0$

$\therefore x = \dfrac{1+\sqrt{5}}{2}$ units only

Q 1 **ACE**

What is the exact value of $\sec 45^\circ$?

(A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{2}}$ (C) 1 (D) $\sqrt{2}$

$\sec 45^\circ = \frac{1}{\cos 45^\circ} = \frac{1}{1/\sqrt{2}} = \sqrt{2}$ 1 Mark: D

Q 2 **AMP**

Find the exact value of $\sin 45^\circ . \cos 30^\circ$ 1

$$\frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2\sqrt{2}} \text{ or } \frac{\sqrt{6}}{4} \checkmark$$

Q 3 **ASCHAM**

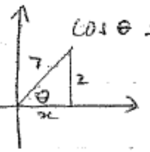
If $\sin\theta = \frac{2}{7}$ and θ is acute, find the exact values of $\cos\theta$. 2

$7^2 = x^2 + 2^2$

$49 = x^2 + 4$

$x^2 = 45$

$x = \sqrt{45}$



$\cos\theta = \frac{\sqrt{45}}{7} = \frac{3\sqrt{5}}{7}$

Q4 **ASCHAM**

Find the exact value of $\sin 225^0$ (1)

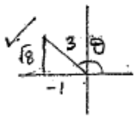
$\sin 225^\circ = -\sin 45^\circ$

$= -\frac{1}{\sqrt{2}}$ (1)

Q5 **ASCHAM**

If $\cos\theta = -\frac{1}{3}$ and $\tan\theta < 0$, find the value of $\sin\theta$ (2)

$\cos\theta = -\frac{1}{3}$ $\tan\theta < 0$ ✓



$\sin\theta = \frac{\sqrt{8}}{3}$ ✓

$\left(\text{or } \frac{2\sqrt{2}}{3}\right)$ (2)

Q6 **BAULKHAM**

Given $\sin\theta = \frac{3}{7}$ and that $\tan\theta < 0$ find the exact value of $\sec\theta$ for $0 \le \theta \le 360°$ without finding θ and showing all working. 3

$\sin\theta > 0 \quad \tan\theta < 0$ 1

$\therefore$ *2nd quad*

$\cos\theta = -\frac{\sqrt{40}}{7}$ 1

$\therefore \sin\theta = -\frac{7}{\sqrt{40}}$ 1

Q7 Simplify $\sec(180^0 - \theta).\tan(90^0 - \theta)$ 3

BAULKHAM

$\sec(180^\circ - \theta) \cdot \tan(90^\circ - \theta)$

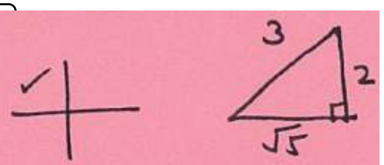
$= \dfrac{1}{\cos(180 - \theta)} \cdot \cot\theta$

$\dfrac{1}{-\cos\theta} \cdot \dfrac{\cos\theta}{\sin\theta}$

$= -\dfrac{1}{\sin\theta}$ or $-\operatorname{cosec}\theta$

Q8 If $90^0 \le \theta \le 180^0$ and $\sin\theta = \dfrac{2}{3}$, find the exact value of $\cos\theta$. 2

BAULKHAM



$\cos\theta = -\sqrt{5}/3$

Q9 Find the exact value of $\tan(-150^\circ)$. 1

WESTERN

$\tan(-150^\circ) = \tan 210^\circ$

$= \tan 30^\circ$

$= \dfrac{1}{\sqrt{3}}$

Q10 Find the exact value of:

(i) $\cos 150^\circ$

(ii) $\tan 300^\circ$

GIRAWEEN

a) i) $\cos 150^\circ = -\cos 30^\circ$

$= -\dfrac{\sqrt{3}}{2}$

ii) $\tan 300^\circ = -\tan 60^\circ$

$= -\sqrt{3}$

Q11 Given θ is in the second quadrant and $\cos\theta = -\dfrac{1}{2}$, find the exact value for $\sin\theta + \tan\theta$ 3

GIRAWEEN

(c) $\cos\theta = -\frac{1}{2}$

$\theta = (180 - 60)^\circ = 120^\circ$

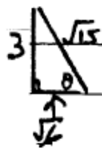
$\sin 120^\circ + \tan 120^\circ$

$= \dfrac{\sqrt{3}}{2} + \left(-\dfrac{\sqrt{3}}{1}\right)$

$= \dfrac{-\sqrt{3}}{2}$ (3)

Q12 Given that $\sin\theta = \dfrac{3}{\sqrt{15}}$ and $\cos\theta < 0$, find the exact value of $\tan\theta$. 2

KINGS



$\tan\theta = -\dfrac{3}{\sqrt{6}}$

Q13 Find the exact value of $\sec(60^\circ)$. **1**

BAULKHAM

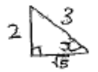
Answer

$$\sec(60^\circ) = \frac{1}{\cos(60^\circ)}$$
$$= \frac{1}{\frac{1}{2}}$$
$$= 2$$

Q14 If $\sin x = \frac{2}{3}$ and $90^0 < x < 180^0$, find the exact value of $\cot x$. **[2]**

SCOTS

i) $\sin x = \frac{2}{3}$



$3^2 - 2^2 = 5$

since $90^0 < x < 180^0$

$\tan x < 0 \therefore \tan x = -\frac{2}{\sqrt{5}}$ (1)

$\therefore \cot x = -\frac{\sqrt{5}}{2}$ (1)

Q15 Find the exact value of $\tan 30^0 + \sec 300^0$ **[2]**

SCOTS

i) $\tan 30 + \sec 300$

$= \frac{1}{\sqrt{3}} + \frac{1}{\cos 300}$

$= \frac{1}{\sqrt{3}} + \frac{1}{\cos 60}$

$= \frac{1}{\sqrt{3}} + \frac{1}{\frac{1}{2}}$ (1)

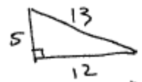
$= \frac{1}{\sqrt{3}} + 2$

Q16 If $\sin \beta = \frac{5}{13}$ and $90° < \beta < 180°$, find the values of $\cos \beta$ and $\tan \beta$. **2**

ST CATH

) $\sin \beta = \frac{5}{13}$

in Q 2



$\cos \beta = -\frac{12}{13}$ $[= -0.9230\ldots]$

$\tan \beta = -\frac{5}{12}$ $[= -0.41\dot{6}]$

Q17 Find the exact value of $\sec 510^0$. **3**

Express your answer with a rational denominator.

GIRAWEEN

$$\sec 510 = \sec(360 + 150)$$
$$= \sec 150$$
$$= \sec(180 - 30)$$
$$= -\sec 30$$
$$= \frac{-2}{\sqrt{3}}$$
$$= \frac{-2}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$
$$= \frac{-2\sqrt{3}}{3}$$

Trigonometry selective Part C: equations

Q1 ACE

If $\sec\theta = 2$ and $0^0 \le \theta \le 360^0$, **2**

Find the value(s) of θ.

$\cos\theta = \frac{1}{2}$

$\therefore\ \theta = 60^0, 300^0$ ✓✓

Q2 ASCHAM

If $\tan\theta = -\sqrt{3}$ for $0^0 \le \theta \le 360^0$, find θ (2)

$\tan\theta = -\sqrt{3}$

(acute $\theta = 60^\circ$)

$\theta = 120^\circ$ or 300° ✓ ✓ (2)

Q3 ASCHAM

Solve for $0^0 \le \theta \le 360^0$

i) $\cos^2\theta - \frac{1}{2} = 0$ **2**

ii) $\sin\theta = \sqrt{3}\cos\theta$ **2**

a) i) $\cos^2\theta - \frac{1}{2} = 0$

$\cos^2\theta = \frac{1}{2}$

$\cos\theta = \pm\frac{1}{\sqrt{2}}$

$\theta = 45^0, 135^0, 225^0, 315^0$ ✓ ✓

$\sin\theta = \sqrt{3}\cos\theta$

$\frac{\sin\theta}{\cos\theta} = \sqrt{3} = \tan\theta$

$\therefore\ \theta = 60^\circ, 240^\circ$ ✓ ✓

Q4 BAULKHAM

Solve for θ where $0^0 \le \theta \le 360^0$.

i) $\sin\theta = \frac{1}{\sqrt{2}}$ 2

ii) $3\tan^3\theta = \tan\theta$ 3

(i) ✓ ✓ $\sin\theta = \frac{1}{\sqrt{2}}$

$\theta = 45^\circ, 135^\circ$ ✓✓

(ii) $\tan\theta(3\tan^2\theta - 1) = 0$

$\therefore \tan\theta = 0$ $\tan\theta = \pm\frac{1}{\sqrt{3}}$ ✓

$\theta = 0^\circ, 180^\circ, 360^\circ$ ✓

$\theta = 30^\circ, 210^\circ, 150^\circ, 330^\circ$ ✓

Q5 BAULKHAM

If $\sin(2x + 20)^\circ = \cos(3x - 80)^\circ$ find x **2**

$2x + 20 + 3x - 80 = 90^\circ$

$5x - 60 = 90$

$x = 30^\circ$ 2

Q6

BAULKHAM

Solve for $0 \leq \theta \leq 360^{\circ}$

$2\cos^2\theta = 2 + \sin\theta$ **3**

$2\cos^2\theta = 2 + \sin\theta$

$2(1 - \sin^2\theta) = 2 + \sin\theta$

$\therefore 2\sin^2\theta + \sin\theta = 0$

$\sin\theta(2\sin\theta + 1) = 0$ 3

$\sin\theta = 0 \qquad \sin\theta = -\frac{1}{2}$

$\therefore \theta = 0, 180^{\circ}, 360^{\circ}, 210^{\circ}, 330^{\circ}$

Q7

BAULKHAM

Solve for $0 \leq \theta \leq 360^{\circ}$

$\sec 2\theta = \cos 2\theta$ **3**

$\sec 2\theta = \cos 2\theta$

$\frac{1}{\cos 2\theta} = \cos 2\theta$

$\cos^2 2\theta = 1$

$\cos 2\theta = \pm 1$ 3

$2\theta = 0, 180, 360, 540, 720$

$\theta = 0, 90, 180, 270, 360$

Q8

CRANBROOK

For $0 \leq x \leq 360^{0}$ solve $2\tan^2\theta - 5\sec\theta + 5 = 0$ **4**

12. Solve for $0 \leq \theta \leq 360^{\circ}$

$2\tan^2\theta - 5\sec\theta + 5 = 0$

$2(\sec^2\theta - 1) - 5\sec\theta + 5 = 0$

$2\sec^2\theta - 5\sec\theta + 3 = 0$

$(2\sec\theta - 3)(\sec\theta - 1) = 0$

$\sec\theta = \frac{3}{2} \qquad \sec\theta = 1$

$\therefore \cos\theta = \frac{2}{3} \qquad \cos\theta = 1$

$\theta = 48^{\circ}11', 311^{\circ}49', 0, 360$

Q9

GIRAWEEN

Solve the following for $0^{o} < \theta < 360^{o}$.

(i) $2\cos\theta + 1 = 0$

(ii) $\tan^2\theta = 3$

i) $2\cos\theta + 1 = 0$

$2\cos\theta = -1$

$\cos\theta = \frac{-1}{2}$

$\therefore \theta = 120^{\circ}, 240^{\circ}$

ii) $\tan^2\theta = 3$

$\tan\theta = \pm\sqrt{3}$

$\theta = 60^{\circ}, 120^{\circ}, 240^{\circ}, 300^{\circ}$

Q10 GOSFORD

Solve $2\cos 2\theta = -1$ for $0° \le \theta \le 180°$ **3**

$\cos 2\theta = -\frac{1}{2}$ $\quad 0° \le 2\theta \le 360°$

$2\theta = 120°, 240°$

$\theta = 60°, 120°$ (3)

Q11 GOSFORD

Solve $2\sin^2 x = 1$ where $-180° \le x \le 180°$ **2**

$2\sin^2 x = 1$

$\sin^2 x = \frac{1}{2}$ (2)

$\sin x = \pm\frac{1}{\sqrt{2}}$

$\therefore x = -135°, -45°, 45°, 135°$

Q12 KINGS

Solve $2\cos^2 x = \sqrt{3}\cos x$ for $-180° \le x° \le 180°$ **3**

$2\cos^2 x - \sqrt{3}\cos x = 0$

$\cos x(2\cos x - \sqrt{3}) = 0$

$\cos x = 0 \qquad \cos x = \frac{\sqrt{3}}{2}$

$\therefore x = \pm 90°, \pm 30°$

① mark ① mark

Q13 KINGS

Solve $(5\sin\theta + 2)(2\cos\theta - 5) = 0$ for $0° \le \theta \le 360°$. Answer to the nearest minute. **3**

(b) $(5\sin\theta + 2)(2\cos\theta - 5) = 0$ for $0 \le \theta \le 360°$

$\sin\theta = -\frac{2}{5}$ or $\cos\theta = \frac{5}{2} = 2\frac{1}{2}$

basic angle, $\theta = 23°35'$ No solutions.

~~S~~	~~A~~
T	C

$\therefore \theta = 180 + 23°35'$ and $360 - 23°35'$

$\theta = 203°35', 336°25'$

Q14 KINGS

What is the solution to the equation $\cos\left(\frac{\theta}{2} + 20°\right) = \sin\theta$ for $0° \le \theta \le 90°$?

(A) $\theta = 20°$

(B) $\theta = 40°$

(C) $\theta = \frac{160°}{3}$

$\boxed{\text{(D) } \theta = \frac{140°}{3}}$

Q15 Solve $4\sin^2\theta=3$ for $-180^\circ \le \theta \le 180^\circ$. **2**

QAT

Answer

$$4\sin^2\theta=3$$
$$\sin^2\theta=\frac{3}{4}$$
$$\sin\theta=\pm\frac{\sqrt{3}}{2}$$
$$\theta=-60^\circ, -120^\circ, 60^\circ, 120^\circ$$

Q16 Solve $2\cos^2 x-3\cos x-2=0$ for $-180^\circ \le x \le 180^\circ$ 3

GOSFORD

b) $2\cos^2 x - 3\cos x - 2 = 0$

$(2\cos x + 1)(\cos x - 2) = 0$

$\cos x = -\frac{1}{2}$ $\quad \cos x = 2$ no sol.

S | A, T | C $\quad$ acute $x = 60^\circ$

$\therefore$ $x = 120^\circ, -120^\circ$

Q17 Find θ if $\sin\theta=\frac{-\sqrt{3}}{2}$ where $0\le\theta\le 360^0$

KINGS

(A) $\theta=60^0$ and 120^0

(B) $\theta=60^0$ and 240^0

(C) $\boxed{\theta=240^0 \text{ and } 300^0}$

(D) $\theta=120^0$ and 240^0

$\sin\theta=\frac{-\sqrt{3}}{2}$, basic angle is 60^0

$\therefore 180^0+60^0=240^0$ and $360^0-60^0=300^0$

Q18 Find x if $\cos(40+x)^0=\sin(2x-19)^0$ **2**

TRIAL ENT

$$\cos(40+x)^0=\sin(2x-19)^0$$
$$\therefore 40+x+2x-19=90$$
$$3x+21=90$$
$$x=23$$

Q19 Solve for x where $0^0\le x\le 360^0$

TRIAL ENT

$2\sin x=\tan x$ **4**

(b)
$$2\sin x=\tan x$$
$$2\sin x=\frac{\sin x}{\cos x}$$
$$2\sin x\cos x=\sin x$$
$$2\sin x\cos x-\sin x=0$$
$$\sin x(2\cos x-1)=0$$
$$\sin x=0 \therefore x=0,180^0,360^0$$
$$\text{or } \cos x=\frac{1}{2}\therefore x=60^0,300^0$$

Q 1 ACE

Prove $\tan\theta + \sec\theta = \dfrac{1+\sin\theta}{\cos\theta}$ **2**

$$\begin{aligned}\text{LHS} &= \tan\theta + \sec\theta \\ &= \frac{\sin\theta}{\cos\theta} + \frac{1}{\cos\theta} \\ &= \frac{1+\sin\theta}{\cos\theta} \\ &= \text{RHS}\end{aligned}$$

Q 2 ACE

Prove $\sin\theta\cos\theta\tan\theta = 1-\cos^2\theta$. **2**

$$\begin{aligned}\text{LHS} &= \sin\theta\cos\theta\tan\theta \\ &= \sin\theta\cos\theta \times \frac{\sin\theta}{\cos\theta} \\ &= \sin^2\theta \\ &= 1-\cos^2\theta \\ &= \text{RHS}\end{aligned}$$

Q 3 AMP

Prove the identity $\dfrac{(\cos\theta+\sin\theta)^2}{\sin\theta\cos\theta} = 2 + \operatorname{cosec}\theta\sec\theta$ **3**

$$\text{L.H.S.} = \frac{\cos^2\theta + \sin^2\theta + 2\cos\theta\sin\theta}{\sin\theta\cos\theta}$$

$$= \frac{1+2\sin\theta\cos\theta}{\sin\theta\cos\theta} \quad \left\{\because\ \cos^2\theta+\sin^2\theta=1\right\} \checkmark$$

$$= \frac{1}{\sin\theta\cos\theta} + \frac{2\sin\theta\cos\theta}{\sin\theta\cos\theta} \checkmark$$

$$\left(\because\ \operatorname{cosec}\theta = \frac{1}{\sin\theta};\ \sec\theta = \frac{1}{\cos\theta}\right) \checkmark$$

$$= \operatorname{cosec}\theta\sec\theta + 2 = \text{R.H.S.}$$

Q4 ASCHAM

Show that $\dfrac{\cos\theta}{1-\sin\theta} - \dfrac{\cos\theta}{1+\sin\theta} = 2\tan\theta$ **2**

$$\begin{aligned}\text{LHS} &= \frac{\cos\theta}{1-\sin\theta} - \frac{\cos\theta}{1+\sin\theta} \\ &= \frac{\cos\theta(1+\sin\theta) - \cos\theta(1-\sin\theta)}{1-\sin^2\theta} \\ &= \frac{\cos\theta + \cos\theta\sin\theta - \cos\theta + \cos\theta\sin\theta}{\cos^2\theta} \\ &= \frac{2\sin\theta\cos\theta}{\cos^2\theta} \\ &= 2\sin\theta = 2\tan\theta = \text{RHS}\end{aligned}$$

Q5 ASCHAM

Simplify $\dfrac{1-\cos^2\theta}{\sin\theta\cos\theta}$ (2)

$$\begin{aligned}\frac{1-\cos^2\theta}{\sin\theta\cos\theta} &= \frac{\sin^2\theta}{\sin\theta\cos\theta} \checkmark \\ &= \frac{\sin\theta}{\cos\theta} \\ &= \tan\theta \checkmark \quad (2)\end{aligned}$$

Q 6 Prove that $\dfrac{\sin\theta}{\cos\theta}+\dfrac{\cos\theta}{\sin\theta}=\sec\theta\,\text{cosec}\,\theta$ **3**

BAULKHAM

$$\begin{aligned}LHS &= \frac{\sin^2\theta+\cos^2\theta}{\cos\theta\sin\theta} & 1\\ &= \frac{1}{\cos\theta\sin\theta} & 1\\ &= \sec\theta\,\text{cosec}\,\theta & 1\\ &= RHS\end{aligned}$$

Q 7 Prove that $\dfrac{1}{\sec x+\tan x}=\sec x-\tan x$ **3**

BAULKHAM

$$\begin{aligned}LHS &= \frac{1}{\sec x\tan x}\times\frac{\sec x-\tan x}{\sec x-\tan x} & 1\\ &= \frac{\sec x-\tan x}{\sec^2 x-\tan^2 x}\\ &= \frac{\sec x-\tan x}{1} & 1\\ &= \sec x-\tan x = RHS & 1\end{aligned}$$

OTHER METHODS

Q 8 Prove $(\cot\theta+\text{cosec}\,\theta)^2=\dfrac{1+\cos\theta}{1-\cos\theta}$ **3**

BAULKHAM

9b) $LHS=(\cot\theta+\text{cosec}\,\theta)^2=$ ✓

$\cot^2\theta+2\dfrac{\cos\theta}{\sin\theta}\cdot\dfrac{1}{\sin\theta}+\dfrac{1}{\sin^2\theta}=\dfrac{\cos^2\theta+2\cos\theta+1}{\sin^2\theta}$ ✓

or different method.

$R.HS=\dfrac{1+\cos\theta}{1-\cos\theta}\cdot\dfrac{1+\cos\theta}{1+\cos\theta}=\dfrac{(1+\cos\theta)^2}{1-\cos^2\theta}$ ✓ $=\dfrac{1+2\cos\theta+\cos^2\theta}{\sin^2\theta}=LHS$

Q 9 Simplify $\dfrac{1}{\text{cosec}^2\theta}+\dfrac{1}{\sec^2\theta}+\dfrac{1}{\cot^2\theta}$.

WESTERN

$$\begin{aligned}&\frac{1}{\text{cosec}^2\theta}+\frac{1}{\sec^2\theta}+\frac{1}{\cot^2\theta}\\ &= \sin^2\theta+\cos^2\theta+\tan^2\theta\\ &= 1+\tan^2\theta\\ &= \sec^2\theta\end{aligned}$$

Q10 Prove $(1-\cos\theta)(1+\sec\theta)=\sin\theta\tan\theta$. **2**

WESTERN

ii) $(1+\cos\theta)(1+\sec\theta)=\sin\theta\tan\theta$

$$\begin{aligned}&LHS\\ &=1+\sec\theta-\cos\theta-\cos\theta\sec\theta\\ &=1+\sec\theta-\cos\theta-\cancel{\cos}\theta\times\frac{1}{\cancel{\cos}\theta}\\ &=\cancel{1}+\sec\theta-\cos\theta-\cancel{1}\\ &=\frac{1}{\cos\theta}-\cos\theta\\ &=\frac{1-\cos^2\theta}{\cos\theta}\\ &=\frac{\sin^2\theta}{\cos\theta}\\ &=\sin\theta\times\frac{\sin\theta}{\cos\theta}\\ &=\sin\theta\tan\theta\\ &=RHS\end{aligned}$$

Q11

GIRAWEEN

Prove the following trigonometric identities

(i) $\tan\theta\operatorname{cosec}\theta = \sec\theta$ **2**

i) $\tan\theta\operatorname{cosec}\theta = \frac{\sin\theta}{\cos\theta}\cdot\frac{1}{\sin\theta}$

$= \frac{1}{\cos\theta}$

$= \sec\theta$. 2

Q12

GIRAWEEN

Prove the following trigonometric identities

$4\sec^2\theta - 3 = 1 + 4\tan^2\theta$ **2**

ii) $4\sec^2\theta - 3 = 4(\tan^2\theta + 1) - 3$

$= 4\tan^2\theta + 4 - 3$

$= 1 + 4\tan^2\theta$ 2

Q13

KINGS

Simplify $(1 - \sin\theta)^2 + \cos^2\theta + 2\sin\theta$. **2**

$1 - \cancel{2\sin\theta} + \sin^2\theta + \cos^2\theta + \cancel{2\sin\theta}$

$= 1 + 1$

$= 2$.

Q14

KINGS

Prove that $(2\tan x + 2)^2 - 4\sec^2 x \equiv 8\tan x$. **2**

$LHS = 4\tan^2 x + 8\tan x + 4 - 4\sec^2 x$

$= 4\tan^2 x + 8\tan x + 4 - 4(1 + \tan^2 x)$

$= \cancel{4\tan^2 x} + 8\tan x + \cancel{4} - \cancel{4} - \cancel{4\tan^2 x}$

$= 8\tan x$

$LHS = RHS$

Q15

KINGS

Simplify $\frac{\sin(90^\circ - \theta)}{\sin(180^\circ - \theta)}$. **2**

(b) $\frac{\sin(90 - \theta)}{\sin(180 - \theta)} = \frac{\cos\theta}{\sin\theta}$ ✓

$= \cot\theta$. ✓

Q16

SCOTS

Prove that $\cot\theta + 2\sec\theta = \frac{1 - \sin^2\theta + 2\sin\theta}{\sin\theta\cos\theta}$ [2]

c) Prove

$\cot\theta + 2\sec\theta = \frac{1 - \sin^2\theta + 2\sin\theta}{\sin\theta\cos\theta}$

$RHS = \frac{\cos^2\theta + 2\sin\theta}{\sin\theta\cos\theta}$

$= \frac{\cos^2\theta}{\sin\theta\cos\theta} + \frac{2\sin\theta}{\sin\theta\cos\theta}$ 1

$= \frac{\cos\theta}{\sin\theta} + \frac{2}{\cos\theta}$

$= \cot\theta + 2\sec\theta$ 1

$= LHS$

Q 1

ACE

(i) Show that $\cot\theta + \tan\theta = \dfrac{1}{\sin\theta\cos\theta}$ **1**

(ii) Hence or otherwise solve the following equation. **1**

$$\frac{\sec\theta}{\cot\theta+\tan\theta} - \frac{1+\cot\theta}{\operatorname{cosec}\theta} = 1 \quad \text{for } 0^\circ \le \theta \le 360^\circ$$

4(d)(i)

$$\begin{aligned} \text{LHS} &= \cot\theta + \tan\theta \\ &= \frac{\cos\theta}{\sin\theta} + \frac{\sin\theta}{\cos\theta} \\ &= \frac{\cos^2\theta + \sin^2\theta}{\sin\theta\cos\theta} \\ &= \frac{1}{\sin\theta\cos\theta} = \text{RHS} \end{aligned}$$

4(d)(ii)

$$\frac{\sec\theta}{\cot\theta+\tan\theta} - \frac{1+\cot\theta}{\operatorname{cosec}\theta} = 1$$

$$\frac{\frac{1}{\cos\theta}}{\frac{1}{\sin\theta\cos\theta}} - \frac{1+\frac{\cos\theta}{\sin\theta}}{\frac{1}{\sin\theta}} = 1$$

$$\begin{aligned} \sin\theta - (\sin\theta + \cos\theta) &= 1 \\ -\cos\theta &= 1 \\ \cos\theta &= -1 \\ \theta &= 180^\circ \end{aligned}$$

Q 2

ACE

(i) Simplify $2\cos^2 x - 2 + 3\sin^2 x$ **1**

(ii) Hence or otherwise, solve $2\cos^2 x - 3 + 3\sin^2 x = 0$ for $0 \le x \le 360^\circ$. **2**

14(b)(i)

$$\begin{aligned} 2\cos^2 x - 2 + 3\sin^2 x &= 2(\sin^2 x + \cos^2 x) + \sin^2 x - 2 \\ &= \sin^2 x \end{aligned}$$

14(b)(ii)

$$\begin{aligned} 2\cos^2 x - 3 + 3\sin^2 x &= 0 \\ (2\cos^2 x - 2 + 3\sin^2 x) - 1 &= 0 \\ \sin^2 x &= 1 \text{ (from (i))} \\ \sin x &= \pm 1 \\ x &= 90^\circ \text{ or } 270^\circ \end{aligned}$$

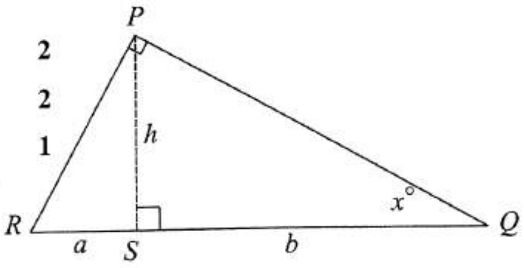
Q 3

ACE

(i) Show that $\angle RPS = x^\circ$ **2**

(ii) Show that $h^2 = ab$ **2**

(iii) Hence find the area of ΔPQR in terms of a and b. **1**



(i)

In ΔPQR

$$\begin{aligned} \angle PQR + \angle RPQ + \angle PRQ &= 180^\circ \quad \text{(Angle sum of triangle is } 180^\circ) \\ x + 90^\circ + \angle PRQ &= 180^\circ \\ \angle PRQ &= 90^\circ - x \end{aligned}$$

In ΔPSR

$$\begin{aligned} \angle RPS + \angle PSR + \angle PRS &= 180^\circ \quad \text{(Angle sum of triangle is } 180^\circ) \\ \angle RPS + 90^\circ + (90^\circ - x^\circ) &= 180^\circ \\ \angle RPS &= x^\circ \end{aligned}$$

4(d)(ii)

In ΔPSR $\tan x^\circ = \dfrac{a}{h}$

In ΔPSQ $\tan x^\circ = \dfrac{h}{b}$

Hence $\dfrac{a}{h} = \dfrac{h}{b}$ (both $\tan x^\circ$)

$h^2 = ab$

4(d)(iii)

$$\begin{aligned} h^2 &= ab \\ h &= \sqrt{ab} \\ A &= \frac{1}{2}bh \\ &= \frac{1}{2} \times (a+b) \times \sqrt{ab} = \frac{(a+b)\sqrt{ab}}{2} \end{aligned}$$

Q 4

BAULKHAM

Express $\dfrac{\sin^3\theta}{\cos\theta} + \sin\theta\cos\theta$ as a single trigonometric ratio **2**

and hence solve $\dfrac{\sin^3\theta}{\cos\theta} + \sin\theta\cos\theta = 1$ for $0 \le \theta \le 360°$

$$\frac{\sin^3\theta}{\cos\theta} + \frac{\sin\theta\cos^2\theta}{\cos\theta}$$

$$= \frac{\sin^2\theta\,(\sin^2\theta + \cos^2\theta)}{\cos\theta}$$

$$= \tan\theta$$

$$\tan\theta = 1$$

$$\theta = 45° \text{ or } 225°$$

Q 5

BAULKHAM

(i) Simplify $\tan^2 A\,(1 - \sin^2 A)$ **2**

(ii) Hence, or otherwise, solve **2**

$4\tan^2 A\,(1 - \sin^2 A) = 1,\ for\ 0° \le A \le 360°$

(i) $\tan^2 A\left(1-\sin^2 A\right) = \dfrac{\sin^2 A}{\cos^2 A}\cos^2 A$

$$= \sin^2 A$$

(ii) $4\tan^2 A\left(1-\sin^2 A\right) = 1$

$$4\sin^2 A = 1$$

$$\sin^2 A = \frac{1}{4}$$

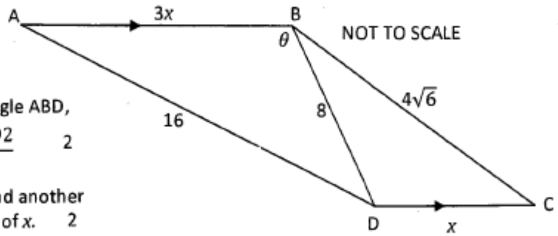
$$\sin A = \pm\frac{1}{2}$$

$$A = 30°,\ 150°,\ 210°,\ 330°$$

Q 6

ASCHAM

In the trapezium ABCD shown in the diagram, AB is parallel to DC, BD = 8cm, AD = 16 cm, BC = $4\sqrt{6}$cm, DC = $x$cm, AB = $3x$cm and $\angle ABD = \theta$.

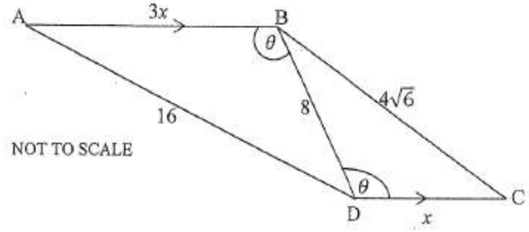


(i) By using the cosine rule in triangle ABD, show that $\cos\theta = \dfrac{9x^2 - 192}{48x}$ **2**

(ii) By considering triangle BCD, find another expression for $\cos\theta$, in terms of x. **2**

(iii) Find the value of x **1**

(iv) Find the value of θ, correct to the nearest minute. **1**



(i) Using cosine rule in triangle ABD

$$\cos\theta = \frac{(3x)^2 + 8^2 - 16^2}{2\times 3x\times 8}$$

$$\cos\theta = \frac{9x^2 + 64 - 256}{48x}$$

$$\cos\theta = \frac{9x^2 - 192}{48x} \qquad \text{———— } 1$$

(ii) In ΔBDC, $\angle BDC = \theta$ (alternate angles, $AB \parallel DC$)

Using cosine rule on triangle BDC

$$\cos\theta = \frac{8^2 + x^2 - \left(4\sqrt{6}\right)^2}{2\times x\times 8}$$

$$\cos\theta = \frac{x^2 + 64 - 96}{16x}$$

$$\therefore\ \cos\theta = \frac{x^2 - 32}{16x} \qquad \text{———— } 2$$

(iii) By considering equations 1 and 2

$$\cos\theta = \frac{9x^2 - 192}{48x} \quad \text{and} \quad \cos\theta = \frac{x^2 - 32}{16x}$$

Let $\cos\theta = \cos\theta$

$$\therefore\ \frac{9x^2 - 192}{48x} = \frac{x^2 - 32}{16x}$$

$$16x\left(9x^2 - 192\right) = 48x\left(x^2 - 32\right)$$

Dividing both sides by $16x$, as $x > 0$

$$9x^2 - 192 = 3(x^2 - 32)$$

$$9x^2 - 192 = 3x^2 - 96$$

$$6x^2 = 96,\ x^2 = 16,\ x = \pm 4$$

But as x is a length, $\therefore\ x = 4$ cm

(iv) Let $x = 4$ in equation 2

$$\cos\theta = \frac{16 - 32}{16\times 4},\quad \cos\theta = \frac{-16}{64},$$

$\theta = 104°29'$ (nearest minute)

Q 7

GIRAWEEN

The diagram shows a square $ABCD$ of side x cm. $PC = 6$ cm, $PB = 2$ cm and $AP = 2\sqrt{5}$ cm.

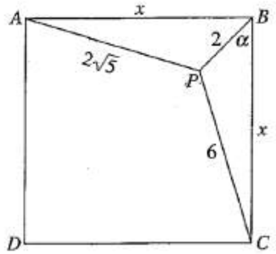
(i) Using the cosine rule in ΔPBC, show that $\cos\alpha = \dfrac{x^2-32}{4x}$ 2

(ii) By considering ΔPBA, show that $\sin\alpha = \dfrac{x^2-16}{4x}$. 2

(iii) Show that the value of x is a solution of

$x^4 - 56x^2 + 640 = 0$ (Hint: $\sin^2\alpha + \cos^2\alpha = 1$). 3

(iv) Solve $x^4 - 56x^2 + 640 = 0$ and determine the length AB of the square. Give reason. 5



(c) (i) In ΔPBC page 6

$\cos\alpha = \dfrac{x^2+2^2-6^2}{2\times x\times 2}$

$= \dfrac{x^2+4-36}{4x}$ 2

$= \dfrac{x^2-32}{4x}$

(ii) In ΔPBA

$\cos(90-\alpha) = \dfrac{x^2+2^2-(2\sqrt{5})^2}{2\times x\times 2}$

$= \dfrac{x^2+2^2-20}{4x}$

$= \dfrac{x^2-16}{4x}$ 2

$\cos(90-\alpha) = \sin\alpha$

$\sin\alpha = \dfrac{x^2-16}{4x}$

(iii) $\sin^2\alpha + \cos^2\alpha = 1$

$\left(\dfrac{x^2-32}{4x}\right)^2 + \left(\dfrac{x^2-16}{4x}\right)^2 = 1$

$\dfrac{x^4-64x^2+1024+x^4-32x^2+256}{16x^2} = 1$

$2x^4 - 96x^2 + 1280 = 16x^2$

$2x^4 - 112x^2 + 1280 = 0$ 3

$x^4 - 56x^2 + 640 = 0$

(iv) $x^4 - 56x^2 + 640 = 0$

Let $u = x^2$

$u^2 - 56u + 640 = 0$

$u = \dfrac{56 \pm \sqrt{3136 - 4\times1\times640}}{2}$

$= \dfrac{56 \pm \sqrt{576}}{2}$

$= \dfrac{56 \pm 24}{2}$

$= 40$ or 16 5

$x^2 = 40$ or $x^2 = 16$

$x = \pm\sqrt{40}$ or $x = \pm 4$

$x = \sqrt{40}$ or $x = 4$

when $x = 4$, ΔPBC does not exist

$\therefore x = \sqrt{40}$ cm

or

$= 2\sqrt{10}$ cm

Q 8

SCEGGS

(i) Show that $\tan\theta + \cot\theta = \dfrac{1}{\sin\theta\cos\theta}$. 1

(ii) Hence, or otherwise, solve $\dfrac{1+\cot\theta}{\operatorname{cosec}\theta} - \dfrac{\sec\theta}{\tan\theta+\cot\theta} = -1$ 3

$0° \le \theta \le 360°$

c) i) LHS $= \tan\theta + \cot\theta$

$= \dfrac{\sin\theta}{\cos\theta} + \dfrac{\cos\theta}{\sin\theta}$

$= \dfrac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta}$ ✓

$= \dfrac{1}{\sin\theta\cos\theta}$

ii) $\dfrac{1+\cot\theta}{\operatorname{cosec}\theta} - \dfrac{\sec\theta}{\tan\theta+\cot\theta}$

$= \dfrac{1+\frac{\cos\theta}{\sin\theta}}{\frac{1}{\sin\theta}} - \dfrac{\frac{1}{\cos\theta}}{\frac{1}{\sin\theta\cos\theta}}$ ✓ from i)

$= \left(1+\dfrac{\cos\theta}{\sin\theta}\right)\times\sin\theta - \dfrac{1}{\cos\theta}(\sin\theta\cos\theta)$

$= \sin\theta + \cos\theta - \sin\theta$

$= \cos\theta$ ✓

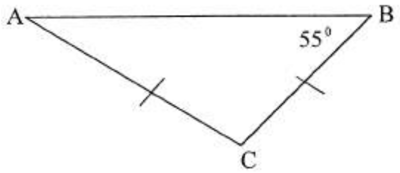
$\therefore \cos\theta = -1$

$\therefore \theta = 180°$ ✓ Reason - 3

Q 9

In the diagram ABC is a triangle where AC = CB, $\angle ABC = 55^0$ and the area of the triangle is $60cm^2$.

TRIAL ENT



(i) Show that $AC = \sqrt{120\operatorname{cosec}70^0}$. **2**

(ii) Show that $AB = \dfrac{\sqrt{120\operatorname{cosec}70^0} \times \sin 70^0}{\sin 55^0}$ **2**

(iii) Find the perimeter of triangle ABC correct to 1 decimal place. **1**

(13)(a)(i)

$\angle ACB = 180 - (55 + 55) = 70^0$

$60 = \frac{1}{2}(AC)^2 \sin 70$

$AC^2 = \dfrac{120}{\sin 70}$

$AC = \sqrt{120\text{cosec}70^0}$

(ii)

$\dfrac{AB}{\sin 70} = \dfrac{AC}{\sin 55}$

$AB = \dfrac{AC\sin 70^0}{\sin 55^0}$

$= \dfrac{\sqrt{120\text{cosec}70^0} \times \sin 70^0}{\sin 55}$

(iii)

$P = 2 \times 11.3005 + 12.9634$

$= 35.56.. = 35.6\text{cm}$

Q10

TRIAL ENT

(i) Show that $\sin^4 3x - \cos^4 3x = 1 - 2\cos^2 3x$ **2**

(ii) Solve $\sin^4 3x - \cos^4 3x = -1$ where $0 \le x \le 360^0$ **3**

(a)(i)

$\sin^4 3x - \cos^4 3x = 1 - 2\cos^2 3x$

LHS

$\sin^4 3x - \cos^4 3x$

$= (\sin^2 3x + \cos^2 3x)(\sin^2 3x - \cos^2 3x)$

$= 1 \times (1 - \cos^2 3x - \cos^2 3x)$

$= 1 - 2\cos^2 3x$

(ii)

$\sin^4 3x - \cos^4 3x = -1$

$1 - 2\cos^2 3x = -1$

$\cos^2 3x = 1$

$\cos 3x = \pm 1$

$3x = 0, 180, 360, 540, 720, 900, 1080$

$x = 0^0, 60^0, 120^0, 180^0, 240^0, 300^0, 360^0$

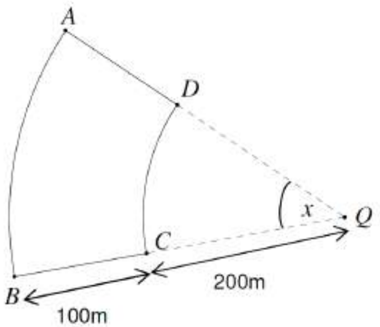
Q 1 ABBOTSLEIGH

In the figure AB and CD are circular arcs which subtend an angle x radians at the centre Q where $0 < x < \pi$ and AQ, BQ are radii. The length BC is 100m and CQ is 200m.

(i) Find expressions in terms of x for the length of the arcs AB and CD. **1**

(ii) A man lives at A and there is a bus stop at B, with the paths AB, BC, CD and DA in the figure forming a road system. For what values of x is it shorter for the man to walk along route $ADCB$ rather than along the arc AB? **2**

(iii) Given $x = \frac{\pi}{5}$ find the area enclosed by the roads linking A, B, C and D. **2**



(a) (i) $l = r\theta$

$\underline{\underline{AB = 300x}}$

$\underline{\underline{CD = 200x}}$

ii) Route $ADCB = 100 + 200x + 100$

$= 200 + 200x$

$200 + 200x < 300x$

$200 < 100x \quad \therefore x > 2$

$\underline{\underline{2 < x < \pi \text{ radians}}}$ (since $0 < x < \pi$ by defn).

(iii) Area $ADCB$ = Area sector AQB − Area sector DQ

$= \frac{1}{2} \times 300^2 \times \left(\frac{\pi}{5}\right) - \frac{1}{2} \times 200^2 \times \frac{\pi}{5}$

$= \frac{\pi}{10}(90000 - 40000)$

$= \underline{\underline{5000\pi}}$ square units

Q 2 ACE

Find the exact value of $\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{2\pi}{3}\right)$ **3**

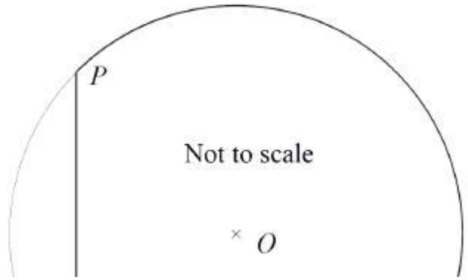
$$\cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{2\pi}{3}\right) = \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2}$$
$$= \frac{\sqrt{2} + \sqrt{3}}{2}$$

Q 3 ACE

A coffee table is the shape of a circle with a small segment removed as shown below. The circle has a centre O and radius 0.5 metres. The length of the straight edge PQ is also 0.5 metres.

(i) Explain why $\angle POQ = \frac{\pi}{3}$ **1**

(ii) Find the area of the coffee table. Answer correct to 2 decimal places. **3**



3(b) (i)

$OP = 0.5$ m (radii)

$OQ = 0.5$ m (radii)

$PQ = 0.5$ m (given)

ΔPOQ is an equilateral triangle. All sides are equal.

All angles in an equilateral triangle are 60°.

$\therefore \angle POQ = \frac{\pi}{3}$

(b) ii)

Area of whole circle $= \pi \times 0.5^2$

$= 0.25\pi$

Area of sector POQ $= \frac{1}{2}r^2\theta$

$= \frac{1}{2} \times 0.5^2 \times \frac{\pi}{3}$

$= \frac{\pi}{24}$

Area of ΔPOQ $= \frac{1}{2}pq\sin O$

$= \frac{1}{2} \times 0.5^2 \times \sin\frac{\pi}{3}$

$= \frac{\sqrt{3}}{16}$

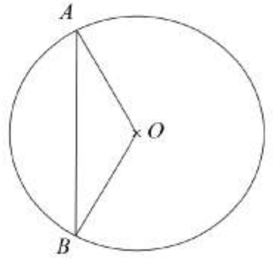
Area of coffee table

$= 0.25\pi - \left(\frac{\pi}{24} - \frac{\sqrt{3}}{16}\right)$

$= 0.762751645\ldots$

$\approx 0.76 \text{ m}^2$

Q 4 ACE



A circle has centre O and radius of 12 cm.
The length of arc AB is 8π cm.

(i) What is the size of $\angle AOB$? Answer in radians. **1**

(ii) Find the area of the minor segment cut off by the chord AB. **2**

13(e) (i)

$$l = r\theta$$
$$8\pi = 12\theta$$
$$\theta = \frac{2\pi}{3}$$

13(e) (ii)

$$A = \frac{1}{2}\times r^2 \times \theta$$
$$= \frac{1}{2}\times 12^2 \times \frac{2\pi}{3}$$
$$= 48\pi \text{ cm}^2$$

$$A = \frac{1}{2}ab\sin O$$
$$= \frac{1}{2}\times 12\times 12\times \sin\frac{2\pi}{3}$$
$$= 72\times\frac{\sqrt{3}}{2} = 36\sqrt{3} \text{ cm}^2$$

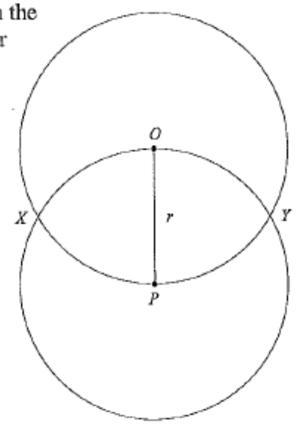
Area of segment $= 48\pi - 36\sqrt{3} \approx 88.44 \text{ cm}^2$

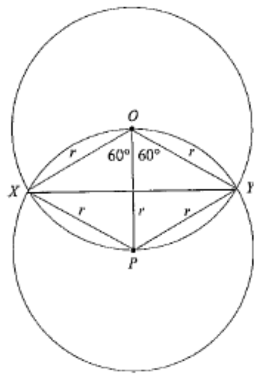
Q 5 ACE

Two equal circles with centres O and P intersect at X and Y as shown in the diagram. The centres of each circle lie on the circumference of the other circle.

(i) Calculate the exact area of the region $XOYP$. **3**

(ii) What fraction of the circle centre O lies outside the region $XOYP$. **2**





10(b) (i)

Area of segment XYP

$$A = \frac{1}{2}r^2(\theta - \sin\theta)$$
$$= \frac{1}{2}r^2\left(\frac{2\pi}{3} - \sin\frac{2\pi}{3}\right)$$
$$= \frac{1}{2}r^2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

Area of the region $XOYP$ is twice the area of segment XYP

$$A = 2\times\frac{1}{2}r^2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$
$$= r^2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$

10(b) (ii)

Area outside the region $XOYP$

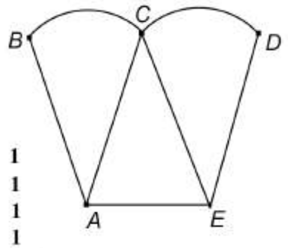
$$A = \pi r^2 - r^2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right)$$
$$= r^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right)$$

Fraction required is $= \dfrac{r^2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{2}\right)}{\pi r^2}$

$$= \frac{1}{3} + \frac{\sqrt{3}}{2\pi}$$

Q 6 — ARC

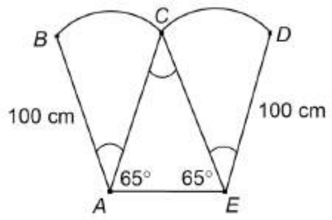
A business logo is designed in the shape below.
ACE is an isosceles triangle with base angles $65°$.
ABC and ECD are sectors of a circle of radius 100 cm.
$\angle BAC = \angle ACE = \angle CED$



NOT TO SCALE

(a) Copy or trace the diagram into your answer booklet and mark the given information on the diagram. **1**
(b) Prove that $\angle ACE = 50°$. **1**
(c) Explain why $BA = AC = CE = ED = 100$ cm. **1**
(d) Find the size of $\angle ACE$ in radians in terms of π. **1**
(e) Find the area ΔACE of in cm^2 correct to two significant figures. **2**
(f) Find the area of sector ABC to two significant figures. Hence find the area of the whole logo and the cost of painting the logo at a rate of \$500/m^2. **3**
(g) Find the cost of gold chain to cover all of the lines and curves in the diagram of the logo at a cost of \$1000/m. **3**

(a)



(b) $\angle ACE + \angle CAE + \angle CEA = 180°$
(angle sum of a triangle)
$\angle ACE + 65° + 65° = 180°$
$\angle ACE = 50°$

(c) BA, AC, CE and ED are all radii of equal circles.

(d) $\angle ACE = \frac{50}{180}\pi$
$= \frac{5\pi}{18}$ radians

(e) $\Delta ACE = \frac{r^2 \sin\theta}{2}$
$= \frac{10\,000 \sin 50°}{2}$
$= 3830.222216...$
$= 3800$ cm^2 (to 2 significant figures)

(f) Area of sector $ABC = \frac{\theta r^2}{2}$
$= \frac{5\pi}{18} \times \frac{10\,000}{2}$
$= 4363.3213...$
$= 4400$ cm^2 (to 2 significant figures)
Area of whole logo $= 4400 \times 2 + 3800 = 12\,600$ cm^2
Cost to paint logo $= \frac{12\,600}{10\,000} \times \$500 = \$630$

(g) Chain required $= 4 \times 100 + 2 \times 100 \times \frac{5\pi}{18} + AE$
$= 400 + 174.5329... + 200\cos 65°$
$= 659.0565...$
$= 659$ cm (to nearest cm)
Cost of chain $= \frac{659}{100} \times \$1000 = \$6590$

$\frac{\frac{AE}{2}}{100} = \cos 65°$
$AE = 200\cos 65°$

Q 7 — ASCHAM

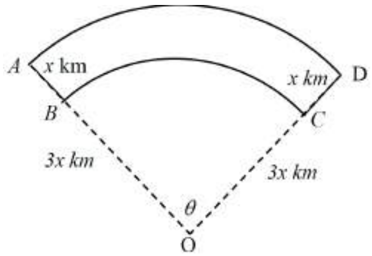
Find the exact value of $\cos 210^0$. **(1)**

$\cos 210° = -\cos 30$
$= -\frac{\sqrt{3}}{2}$

Q 8 — BARKER

Four towns A, B, C and D are joined by roads that are either straight or arcs of concentric circles with centre at O. Town B and C are $3x$ km from O. Towns A and D are both x km from B and C respectively. $\angle AOD = \theta$ radians as shown in the diagram below.

(i) Write an expression in terms of x and θ for the length of arc AD **1**
(ii) A salesperson wants to travel from Town A to Town D but must visit town B and C on the way. Write an expression in terms of θ and x, for the length of this journey from Town A to Town D. **1**
(iii) Find the value of θ for which the journeys in (i) and (ii) are the same distance. **1**



e) i) $AD = 4x\theta$

ii) $ABCD = x + 3x\theta + x$
$= 2x + 3x\theta$

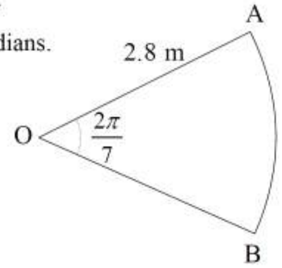
iii) $AD = ABCD$
$4x\theta = 2x + 3x\theta$
$x\theta = 2x$
$x(\theta - 2) = 0$
$\therefore x \neq 0 \quad \theta = 2$ radians

Q 9 BARKER

In the diagram, AB is an arc of a circle with centre O. **2**

The radius OA is 2.8 m and the angle AOB is $\frac{2\pi}{7}$ radians.

Find the area of the sector AOB.



$A = \frac{1}{2} r^2 \theta$

$= \frac{1}{\cancel{2}} \times 2 \cdot 8^2 \times \frac{\cancel{2}\pi}{7}$

$= 3 \cdot 52 \ m^2$ OR $1 \cdot 12 \pi \ m^2$

Q10 BARKER

Find the exact value of $\cos\frac{\pi}{4} + \sin\frac{5\pi}{6}$. **2**

$\cos\frac{\pi}{4} + \sin\frac{5\pi}{6} = \frac{1}{\sqrt{2}} + \sin\frac{\pi}{6}$

$= \frac{1}{\sqrt{2}} + \frac{1}{2}$

$= \frac{\sqrt{2}}{2} + \frac{1}{2}$

$= \frac{1+\sqrt{2}}{2}$

Q11 BARKER

Find the exact value of $\operatorname{cosec}\left(\frac{5\pi}{3}\right)$. **2**

(d) $\operatorname{cosec}\left(\frac{5\pi}{3}\right)$

$= \frac{1}{\sin\left(\frac{5\pi}{3}\right)}$

$= \frac{1}{-\sin\frac{\pi}{3}}$

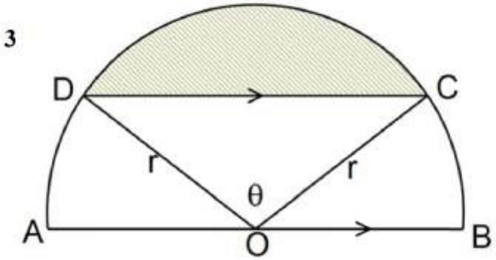
$= \frac{1}{-\frac{\sqrt{3}}{2}}$

$= -\frac{2}{\sqrt{3}}$

Q12 BAULKHAM

The shaded segment has an area one half of the semicircle with centre O and radius r. $CD \parallel AB$.

Prove that $\frac{\pi}{2} = \theta - \sin\theta$. **3**



Ar of Seg CDE $= \frac{1}{4}\pi r^2$ —— (1)

Ar of Δ DOC $= \frac{1}{2} r^2 \sin\theta$ —— (1)

Ar of sector DOC $= \frac{1}{2} r^2 \theta$

$\therefore$ Ar of Seg CDE

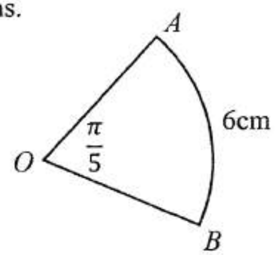
= Ar of sector − Ar of Δ

$\frac{1}{4}\pi r^2 = \frac{1}{2} r^2\theta - \frac{1}{2} r^2 \sin\theta$] (1)

$\frac{1}{2}\pi = \theta - \sin\theta$

Q13 BAULKHAM

The sector shown has arc length of 6 cm and $\angle AOB$ is $\frac{\pi}{5}$ radians.



(i) Find the radius of the circle in terms of π. **1**

(ii) Hence, find the exact area of the sector. **1**

(i) $l = r\theta \quad \therefore r = \frac{6}{\frac{\pi}{5}} = \frac{30}{\pi}$ (1) (9.549) ✓

(ii) $A = \frac{1}{2}r^2\theta$

$= \frac{1}{2}\cdot\left(\frac{30}{\pi}\right)^2\cdot\frac{\pi}{5}$

$= \frac{900\pi}{10\pi^2} = \frac{90}{\pi}$ (1) $(28.648\ldots)$ ✓

Q14 BAULKHAM

What is the exact value of $\cos\frac{7\pi}{6}$

(A) $\frac{\sqrt{3}}{2}$ $\quad$ $\boxed{\text{(B) } -\frac{\sqrt{3}}{2}}$ $\quad$ (C) $\frac{1}{2}$ $\quad$ (D) $-\frac{1}{2}$

Q15 BAULKHAM

The $\lim\limits_{\theta \to 0} \dfrac{\sin\frac{\theta}{4}}{\theta} =$

(A) 1 $\quad$ (B) 4 $\quad$ $\boxed{\text{(C) } \frac{1}{4}}$ $\quad$ (D) 0

Q16 BAULKHAM

Express 63° in radians to 3 significant figures **2**

$63^\circ \times \frac{\pi}{180^\circ} = 1.09955\ldots$ ✓

$= 1.10$ (3 sig fig) ✓

Q17 BAULKHAM

An arc length 7 units subtends an angle θ at the centre of a circle of radius 3 units. **2**
Find the value of θ to the nearest degree.

$3\theta = 7$

$\theta = \frac{7}{3}$ radians ✓

$\theta = 134^\circ$ ✓

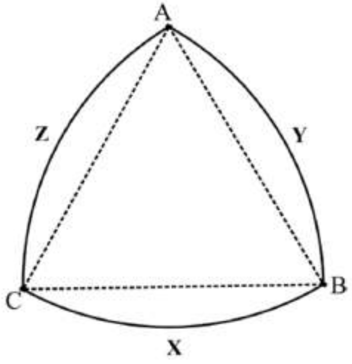
Q18 CARINGBAH

A cam in a lock, AYBXCZ, is drawn below.

ABC is an equilateral triangle of side length 12cm. The arcs AYB, BXC and CZA have centres C, A and B respectively.

(i) Find the perimeter of the cam. **1**

(ii) Find the exact area of the cam. **2**



f) i) $l = r\theta$ $\qquad P = 3l$

$= 12 \times \frac{\pi}{3}$ $\qquad = 12\pi$

ii) $A = \frac{1}{2}r^2\theta + 2\left(\frac{1}{2}r^2(\theta - \sin\theta)\right)$

$= \frac{1}{2} \times 12^2 \times \frac{\pi}{3} + 2\left(\frac{1}{2} \times 12^2\left(\frac{\pi}{3} - \sin\frac{\pi}{3}\right)\right)$

$= 24\pi + \frac{144\pi}{3} - 144 \times \frac{\sqrt{3}}{2}$

$= 72\pi - 72\sqrt{3}$

Q19 CARINGBAH

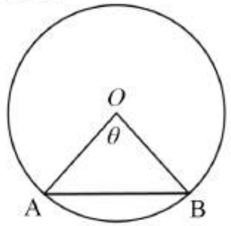
The chord AB divides the area of the circle of radius r into two parts. AB subtends an angle of θ at the centre, as shown.

i) Show that the ratio of areas of major to minor segment is given by **3**

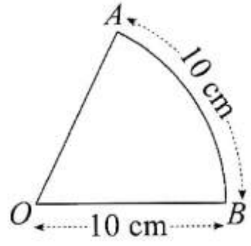
$$\frac{2\pi - \theta + \sin\theta}{\theta - \sin\theta}$$

ii) The chord AB divides the area of the circle in the ratio 15 : 7. **2**

Using $\pi = \frac{22}{7}$, prove that $\theta = 2 + \sin\theta$.



Q20 CARINGBAH



AB is a circular arc, centre O, length 10 cm. **2**

The radius is also 10 cm.

Find the area of sector AOB.

Length of arc AB $= r\theta$

$10 = 10\theta$

$\theta = 1$

Area of sector $= \frac{1}{2}r^2\theta$

$= \frac{1}{2} \times 10^2 \times 1$

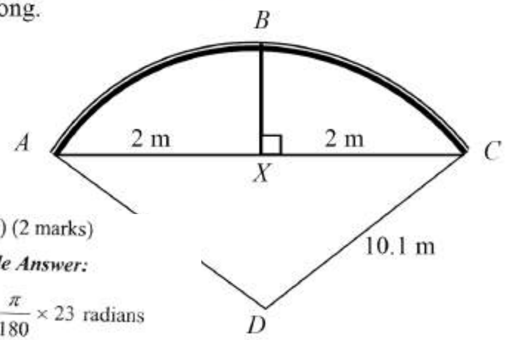
$= 50$ units^2

Q21 CSSA

A bridge's steel arch ABC is part of a circle of radius 10.1 metres. BX bisects the chord AC which is 4 metres long.

(i) Find the size of angle ADC correct to the nearest degree. **2**

(ii) Find the length of steel needed to make the arch ABC. **2**



(b) (i) (2 marks)

Sample Answer:

$\cos\angle ADC = \frac{10.1^2 + 10.1^2 - 4^2}{2\times10.1\times10.1}$

$= 0.92$

$\angle ADC = 23^\circ$

(b) (ii) (2 marks)

Sample Answer:

$23^\circ = \frac{\pi}{180} \times 23$ radians

$= 0.4$ radians

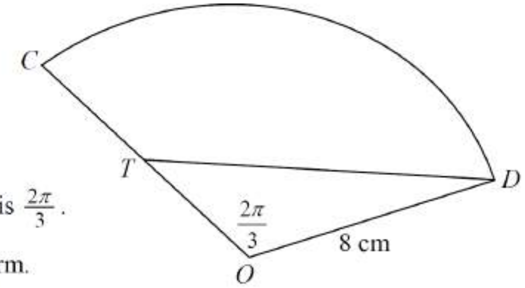
$l = r\theta$

$= 10.1 \times 0.4$

$= 4.04$ metres

Q22 CSSA

In the diagram, CD is an arc of a circle with radius 8 cm and centre O. 3



T is the midpoint of OC. Angle COD is $\frac{2\pi}{3}$.

Find the perimeter of CTD in exact form.

(c) (3 marks)
Sample Answer

Length DT

$OT = 4$ cm (midpoint of OC)

$DT^2 = 4^2 + 8^2 - 2\times4\times8\cos\left(\frac{2\pi}{3}\right)$

$DT^2 = 112$

$DT = 4\sqrt{7}$

Arc length CD

$l = r\theta$

$CD = 8 \times \frac{2\pi}{3}$

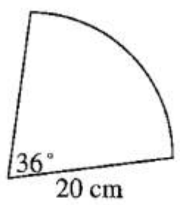
$= \frac{16\pi}{3}$

Perimeter $= DT +$ arc CD $+ CT$

$= \left(4\sqrt{7} + \frac{16\pi}{3} + 4\right)$ cm

Q23 CSSA

What is the perimeter, P, of the sector below with angle 36° and radius 20cm?



(A) $P = 0.5 \times 400 \times \left(\frac{\pi}{5} - \sin\frac{\pi}{5}\right)$ cm

(B) $P = \left(0.5\times400\times\frac{\pi}{5}\right)$ cm

(C) $P = (40+36^\circ)$ cm

(D) $\boxed{P = (40+4\pi) \text{ cm}}$

Q24 CSSA

Find the exact value of $\tan\frac{2\pi}{3}$.

$$\tan\frac{2\pi}{3} = \tan 120^\circ$$
$$= -\tan 60^\circ$$
$$= -\sqrt{3}$$

Q25 CSSA

The area of the sector of a circle with radius 8 cm is $\frac{56\pi}{5}$ cm^2. 2

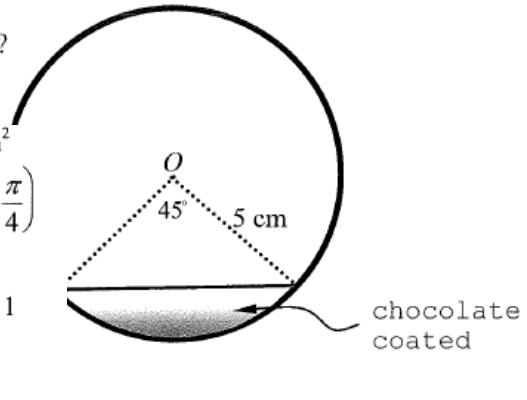
Find the angle that is subtended at the centre of the sector.

$$\frac{56\pi}{5} = \frac{1}{2}\times 8^2 \times\theta$$
$$\frac{56\times2\times\pi}{5\times64} = \theta$$
$$\frac{112\pi}{320} = \theta$$
$$\frac{7\pi}{20} = \theta \text{ or } 63^\circ$$

Q26 CSSA

A circular wafer slice, to be used for a dessert, is to have a portion of its front coated in chocolate as shown in the diagram. The radius of the wafer slice is 5cm. 3

What area of the wafer will remain uncoated?
Give your answer correct to the nearest cm^2.



Area of circular wafer $= \pi\times5^2 = 78.5398\ldots\text{cm}^2$

Area of chocolate segment $= \frac{1}{2}\times5^2\times\left(\frac{\pi}{4} - \sin\frac{\pi}{4}\right)$

$= 0.9786\ldots\text{cm}^2$

$\therefore$ Area uncoated $= 78.5398 - 0.9786 = 77.5611$

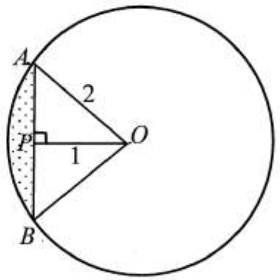
≈ 78 cm^2

Q27 CSSA

The diagram shows a circle with centre O, radius 2cm and $OP = 1$ cm.

(i) Show that the area of the sector AOB is $\frac{4\pi}{3}$ cm^2. **2**

(ii) Hence, or otherwise, find the shaded area. **2**



(d) (i) (2 marks)

Sample answer:

Let $\angle AOP = \theta$

In ΔAOP $\cos\theta = \frac{1}{2}$ $\quad \therefore \theta = \frac{\pi}{3}$

$\therefore \angle AOB = \frac{2\pi}{3}$

$\therefore$ area of the sector $AOB = \frac{1}{2}\times 2^2 \times \frac{2\pi}{3} = \frac{4\pi}{3}$

(d) (ii) (2 marks)

Sample answer:

Shaded area $= \frac{1}{2}\times 2^2\left(\frac{2\pi}{3} - \sin\frac{2\pi}{3}\right)$

$= 2\left(\frac{2\pi}{3} - \frac{\sqrt{3}}{2}\right) = \left(\frac{4\pi}{3} - \sqrt{3}\right) \approx 2.46$

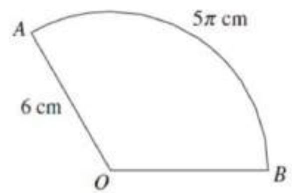
Q28 FORT ST

If $f(x) = 2\sin 3x$ find the exact value of $f'\left(\frac{\pi}{18}\right)$ **2**

$$f(x) = 2\sin 3x$$
$$f'(x) = 6\cos 3x \text{ ❶}$$
$$f'\left(\frac{\pi}{18}\right) = 6\cos\frac{3\pi}{18}$$
$$= 6\cos\frac{\pi}{6}$$
$$= 6\times\frac{\sqrt{3}}{2}$$
$$= 3\sqrt{3} \text{ ❶}$$

Q29 FORT ST

The diagram shows sector AOB.



NOT TO SCALE

What is the exact area of the minor segment cut off by a chord AB?

(A) 15π

(B) $15\pi - 9$

(C) 12π

(D) $12\pi - 3\sqrt{3}$

(B) $15\pi - 9$

Q30 INDEPENDENT

Which expression is a simplification of $\dfrac{\sin(\pi - x)}{\sin\left(\frac{\pi}{2} - x\right)}$?

(A) $\dfrac{-\sin x}{1-\sin x}$

(B) $-\tan x$

(C) $\boxed{\tan x}$

(D) $\dfrac{2(\pi - x)}{\pi - 2x}$

Q31 INDEPENDENT

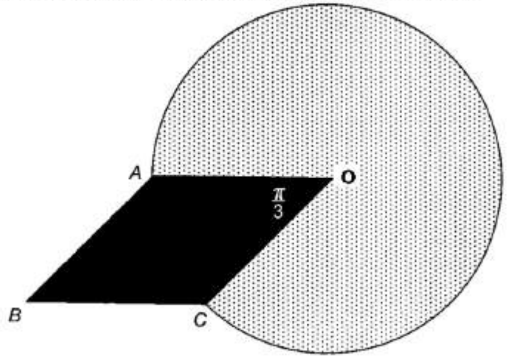
A concrete viewing platform is to be built at a mountain lookout. The platform is formed from a rhombus $AOCB$ with side $AO = 5$m and $\angle AOC = \frac{\pi}{3}$, and the major sector of a circle centre O, radius AO. The concrete is 200mm thick.

The platform is illustrated in the diagram below.

(i) Show that reflex $\angle AOC = \frac{5\pi}{3}$. **1**

(ii) Calculate the area of the platform. **3**

(iii) Find the volume of concrete used to make the platform. **1**



(b)(i)	Reflex $\angle AOB$ = Circle - Acute $\angle AOB = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$ (1 mark)
(b)(ii)	Area = Rhombus + Major Sector AOB = 2 x ΔAOB $+\frac{1}{2}r^2\theta$ (1 mark) $=2\times\frac{1}{2}\times5\times5\times\sin\frac{\pi}{3} + \frac{1}{2}\times5^2\,\frac{5\pi}{3}$ (1 mark) $=\frac{25\sqrt{3}}{2} + \frac{125\pi}{6}$ ≈ 87.1 *metres*^{2} (1 mark)
(b)(iii)	Vol = Area x Thickness $=\left(\frac{25\sqrt{3}}{2} + \frac{125\pi}{6}\right)\times 0.2$ $=5\left(\frac{\sqrt{3}}{2}+\frac{5\pi}{6}\right)$ *metres*^{3} (1 mark or approx. value 1 mark) ≈ 17.42 *metres*^{3}

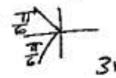
Q 1 ABBOTSLEIGH

Solve $\cos\theta = \frac{-\sqrt{3}}{2}$ for $0 \le \theta \le 2\pi$ **2**

(a) $\cos\theta = -\frac{\sqrt{3}}{2}$

$0 \le \theta \le 2\pi$

Basic angle $= \frac{\pi}{6}$



2nd + 3rd qua

$\theta = \pi - \frac{\pi}{6}, \ \pi + \frac{\pi}{6}$

$\theta = \frac{5\pi}{6}, \ \frac{7\pi}{6}$

Q 2 ACE

Which expression is the simplest possible form of $\frac{\operatorname{cosec}\theta\sec\theta}{\tan\theta}$?

(A) $\sec^2\theta$

(B) $\boxed{\operatorname{cosec}^2\theta}$

(C) $\cot^2\theta$

(D) $\sin^2\theta$

$$\frac{\operatorname{cosec}\theta\sec\theta}{\tan\theta} = \operatorname{cosec}\theta\sec\theta \times \frac{\cos\theta}{\sin\theta}$$
$$= \frac{1}{\sin\theta} \times \frac{1}{\cos\theta} \times \frac{\cos\theta}{\sin\theta}$$
$$= \frac{1}{\sin^2\theta} = \operatorname{cosec}^2\theta$$

Q 3 ACE

What is solution to the equation $\frac{\cos\theta}{\sqrt{3}} = -\frac{1}{2}$ for $0^\circ \le \theta \le 360^\circ$?

(A) $\theta = 30^\circ$ or 330°

(B) $\theta = 60^\circ$ or 300°

(C) $\boxed{\theta = 150^\circ \text{ or } 210^\circ}$

(D) $\theta = 120^\circ$ or 240°

$$\cos\theta = -\frac{\sqrt{3}}{2}$$
$$\theta = 150^\circ \text{ or } 210^\circ$$

Q 4 ACE

Solve the equation $(\cos x + 2)(2\cos x + 1) = 0$ in the domain $0 \le x \le 2\pi$. **2**

$2\cos x + 1 = 0$

$\cos x = -\frac{1}{2}$ or $x = \frac{\pi}{3}$

$\cos x + 2 = 0$

$\cos x = -2$

No solution

In domain $0 \le x \le 2\pi$ the solution is $x = \frac{2\pi}{3}, \frac{4\pi}{3}$

Q 5 ACE

(i) Express $\sin\theta\cos\theta+\dfrac{\cos^3\theta}{\sin\theta}$ as a single trigonometric ratio. 1

(ii) Hence solve $\sin\theta\cos\theta+\dfrac{\cos^3\theta}{\sin\theta}=1$ for $0\le\theta\le 2\pi$. 1

$$\sin\theta\cos\theta+\frac{\cos^3\theta}{\sin\theta}=\frac{\cos\theta}{\sin\theta}\left(\sin^2\theta+\cos^2\theta\right)$$
$$=\cot\theta$$

$\cot\theta=1$

$\theta=\dfrac{\pi}{4}$ or $\dfrac{5\pi}{4}$

Q 6 ACE

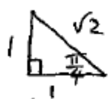
Solve the equation $\sin^2 x=\cos^2 x,\ 0\le x\le\pi$. 2

$\sin^2 x=\cos^2 x \quad 0\le x\le\pi.$ 2

$\tan^2 x=1$

$\tan x=\pm 1$ ✓

$x=\dfrac{\pi}{4}$ or $\dfrac{3\pi}{4}$ ✓



Q 7 BARKER

Solve the equation: 3

$2\cos^2 x=2\sin x\cos x$ where $0\le x\le 2\pi$

o) $2\cos^2 x-2\sin x\cos x=0$

$2\cos x(\cos x-\sin x)=0$

$\cos x=0 \qquad \cos x=\sin x$

$\tan x=1$

$x=\pi/2,\ 3\pi/2 \qquad x=\pi/4,\ 5\pi/4$

$x=\pi/4,\ \pi/2,\ 5\pi/4,\ 3\pi/2$

Q 8 BAULKHAM

Find the value of $\tan x$ when $\tan^2 x+\sec^2 x=9$ 2

$\tan^2 x+1+\tan^2 x=9$ — 1

$2\tan^2 x=8$

$\tan x=\pm 2$ — 1

Q 9 BAULKHAM

Solve $2\sin\theta + 1 = 0$ for $0 \le \theta \le 2\pi$ **2**

$\sin\theta = -\frac{1}{2}$

$\theta = \sin^{-1}(-\frac{1}{2})$

$= \frac{7\pi}{6}, \frac{11\pi}{6}$

Q10 BAULKHAM

Solve for x for $0 \le x \le 2\pi$ $\quad 2\sin^2 x = 3(\cos x + 1)$ **3**

$2(1-\cos^2 x) = 3\cos x + 3$

$2\cos^2 x + 3\cos x + 1 = 0$

$(2\cos x + 1)(\cos x + 1) = 0$

$\cos x = -\frac{1}{2}$ or $\cos x = -1$

Acute $\angle = \frac{\pi}{3}$

$\therefore x = \frac{2\pi}{3}, \frac{4\pi}{3}$ or π

Q11 CARINGBAH

Solve $2\sin\left(x - \frac{\pi}{4}\right) + 1 = 0$ for $0 \le x \le 2\pi$. **2**

Leaving your answer(s) in exact form.

$\sin(x - \frac{\pi}{4}) = -\frac{1}{2}$

$x - \frac{\pi}{4} = \frac{7\pi}{6}$ $\quad x - \frac{\pi}{4} = \frac{11\pi}{6}$

$x = \frac{17\pi}{12}$ $\quad x = \frac{25\pi}{12} = \frac{\pi}{12}$

Q12 CARINGBAH

Solve $4\cos^2\theta - 3 = 0$ for $0 \le \theta \le 2\pi$ **3**

$4\cos^2\theta = 3$

$\cos^2\theta = \frac{3}{4}$

$\cos\theta = \pm\frac{\sqrt{3}}{2}$

$\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$

Q13 CSSA

Solve $2\sin^2\alpha - \sin\alpha = 0$ for $0 < \alpha < 2\pi$. **3**

$$2\sin^2\alpha - \sin\alpha = 0$$
$$\sin\alpha(2\sin\alpha - 1) = 0$$
$$\sin\alpha = 0 \text{ or } \sin\alpha = \frac{1}{2}$$
$$\alpha = \sin^{-1}(0) \text{ or } \alpha = \sin^{-1}\left(\frac{1}{2}\right)$$
$$\text{Acute } \alpha = 0, \frac{\pi}{6}$$
$$\therefore \alpha = \pi, \frac{\pi}{6}, \frac{5\pi}{6}$$

Q14 CSSA

The solution of $\sqrt{2}\cos x - 1 = 0$, where $0 \le x \le 2\pi$ is:

(A) $\frac{\pi}{4}, \frac{5\pi}{4}$ (B) $\boxed{\frac{\pi}{4}, \frac{7\pi}{4}}$ (C) $\frac{\pi}{3}, \frac{2\pi}{3}$ (D) $\frac{\pi}{3}, \frac{5\pi}{3}$

Q15 FORT ST

Solve $2\cos^2 x - 1 = 0$ for $-\pi \le x \le \pi$ **3**

$$2\cos^2 x - 1 = 0$$
$$2\cos^2 x = 1$$
$$\cos^2 x = \frac{1}{2}$$
$$\cos x = \pm\frac{1}{\sqrt{2}}$$
$$\text{For } -\pi \le x \le \pi \quad x = \frac{3\pi}{4}, \frac{\pi}{4}, \frac{-\pi}{4}, \frac{-3\pi}{4}$$

❶ *values* ❶ *signs* ❶ *solutions*

Q16 FRENSHAM

Solve $\sqrt{3}\cos x = \sin x$, $0 \le x \le 2\pi$ **2**

$$\sqrt{3}\cos x = \sin x$$
$$\sqrt{3} = \frac{\sin x}{\cos x}$$
$$\tan x = \sqrt{3}$$
$$x = \frac{\pi}{3}$$

tan positive 1st, 3rd quadrant

$$\therefore \quad x = \frac{\pi}{3}, \frac{4\pi}{3}$$

Q 1

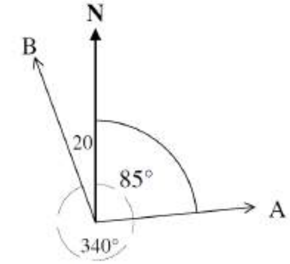
Ashleigh and Bianca set out on a bike ride from point O at the same time. Ashleigh travels at 20 km/h along a straight road in the direction 085° T. Bianca travels at 25 km/h along another straight road in the direction 340° T.

Draw a diagram to represent this information.

(i) Show that $\angle AOB$ is 105° where $\angle AOB$ is the angle between the directions taken by Ashleigh and Bianca. **1**

(ii) Find the distance Ashleigh and Bianca are apart to the nearest kilometre after two hours. **3**

(c)(i)



$$\angle AOB = 85^\circ + 20^\circ$$
$$= 105^\circ$$

12(c)(ii) After 2 hours Ashleigh travels 40 km and Bianca travels 50 km.

$$AB^2 = 40^2 + 50^2 - 2\times 40\times 50\times \cos 105^\circ$$
$$AB^2 = 5135.27618...$$
$$AB = 71.66084133...$$
$$AB \approx 72 \text{ km}$$

Ashleigh and Bianca are 72 km apart after 2 hours.

Q 2

A ship leaves port P and travels on a bearing of 104° a distance of 300 km to point Q. It then turns and travels on a bearing of 239° for 200 km to point R.

(i) Show that $\angle PQR = 45^\circ$. **1**

(ii) What is the distance from R to P? Answer to the nearest kilometre. **2**

(iii) Find the bearing of R from P? Answer to the nearest degree. **2**

Solution is $x = 2$ and $y = -1$.

13(d)(i)

$\angle SPQ + \angle TQP = 180^\circ$ (Cointerior angles, supplementary)

$104^\circ + \angle TQP = 180^\circ$

$\angle TQP = 76^\circ$

$\angle PQR + \angle TQP + \angle TQR = 360^\circ$ (Revolution is 360°)

$\angle PQR + 76^\circ + 239^\circ = 180^\circ$

$\angle PQR = 45^\circ$

13(d)(ii)

$$PR^2 = PQ^2 + QR^2 - 2\times PQ\times QR\times \cos 45^\circ$$
$$= 300^2 + 200^2 - 2\times 300\times 200\times \cos 45^\circ$$
$$PR = 212.4786725...$$
$$\approx 212 \text{ km}$$

Therefore, the distance from R to P is 212 km

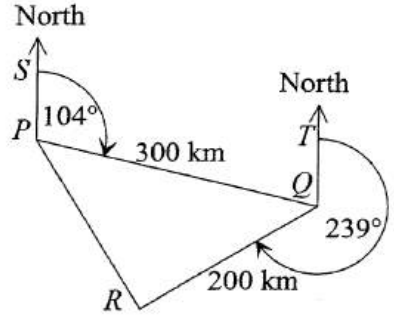
13(d)(iii)

$$\frac{\sin\angle QPR}{200} = \frac{\sin 45^\circ}{212.4786...}$$
$$\sin\angle QPR = \frac{200\sin 45^\circ}{212.4786...}$$
$$\angle QPR = 41.726765...$$
$$\approx 42^\circ$$

Bearing $= 104^\circ + 42^\circ$

$= 146^\circ$

Therefore, the bearing of R from P is 146°



Q3 Two ships leave port A at the same time. Ship X travels on a bearing of 070° and ship Y travels for 15.7 km on a bearing of 98° until the bearing of ship X from ship Y is 300°.

(i) Show that $\angle ABC = 130^\circ$ **1**

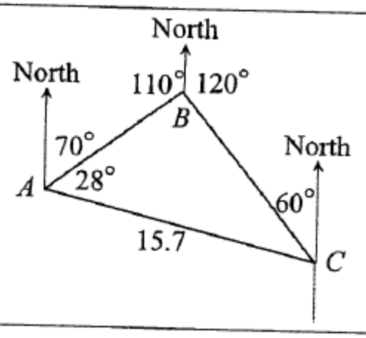
(ii) What is the bearing of port A from ship X? Answer the nearest degree **1**

(iii) How far has ship X travelled? Answer correct to one decimal place. **2**

13(a) (i)

$\angle ABC + 110^\circ + 120^\circ = 360^\circ$

$\angle ABC = 130^\circ$



13(a) (ii)

Bearing $= 120^\circ + 130^\circ$

$= 250^\circ$

The bearing of B from A is 250°

3(a) (iii)

To find the distance AB we require $\angle BCA$

$\angle BCA + 28^\circ + 130^\circ = 180^\circ$

$\angle BCA = 22^\circ$

$$\frac{AB}{\sin 22^\circ} = \frac{15.7}{\sin 130^\circ}$$

$$AB = \frac{15.7 \sin 22^\circ}{\sin 130^\circ}$$

$= 7.67752259\ldots \approx 7.7$ km

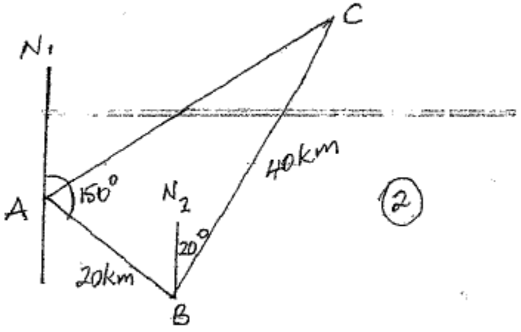
Ship X has travelled 7.7 km

Q4 Two geologists on a large flat mining claim drive 20 km from point A on a bearing of 150^0 T to point B. They then drive 40 km on a bearing of 020^0 T to point C.

i) Show that $\angle ABC = 50^0$ (2)

ii) Find the distance of point C from point A to the nearest kilometre. (2)

b) i)



(2)

ii) $\angle N_2BA = 30^\circ$ (co-int ∠s $N_1A \parallel N_2B$)

$\therefore \angle ABC = 30^\circ + 20^\circ$

$= 50^\circ$ (2)

iii) $AC^2 = 20^2 + 40^2 - 2 \times 20 \times 40 \cos 50$

$= 971.5398\ldots$

$AC = 31.16\ldots$

$\therefore$ distance of C from A is 31 km (to n km) (2)

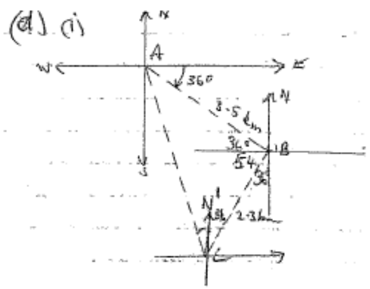
Q5 Mary walks from point A to point B for 3.5 km on a bearing of 126^0. She then changes direction and walks to point C, 2.3 km away on a bearing of 216^0, and stops.

i) Draw a diagram showing all details of her journey. 1

ii) Find the distance AC. Give your answer correct to one decimal place. 2

iii) What is the bearing of point A from point C? Give your answer to the nearest minute. 2

(d) (i)



(ii)
$\angle ABC = 90^0$

$\therefore AC^2 = 3.5^2 + 2.3^2$

$AC = 4.2$ km. (1 d.p.

(iii) Need to find $N'CA$

$\tan A = \frac{2.3}{3.5}$

$A = 33^0 19'$ (nearest minute)

$\therefore N'CA = 180 - (90 + 36 + 33^0 19')$
$= 20^0 41'$

$\therefore$ Bearing is $360^0 - 20^0 41$

=339'19'

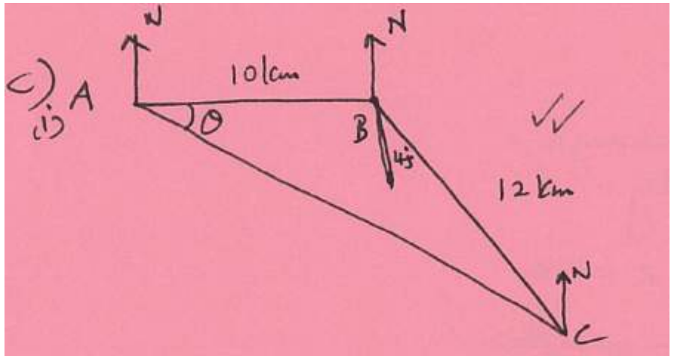
Q6 A person walks 10 km East from A to B and then 12km South East to C.

i) Draw a neat diagram and mark the given information 2

ii) Find the distance to the nearest km between A and C. 2

iii) What is the bearing of A from C? 2

c) (i)



(ii) $AC^2 = 10^2 + 12^2 - 2 \times 10 \times 12 \cos 135^\circ$

$AC \doteqdot 20$ km.

(iii) $\frac{12}{\sin\theta} = \frac{20}{\sin 135^\circ}$

$\sin\theta = \frac{12 \times \sin 135^\circ}{20}$

$= 0.41717$

$\theta = 24^0 39'$

(13)

Bearing N $65^0 21'$ W

$291^0 39'$ T

Q7 Two friends drive 30km from A on a bearing of 150°T to point B. They then drive from B to C, a further 50km on a bearing of 050°T.

i) Show that $\angle ABC = 80^0$ **1**

ii) Find the distance from A to C **2**

iii) Find the bearing of A from C **2**
Show all the working

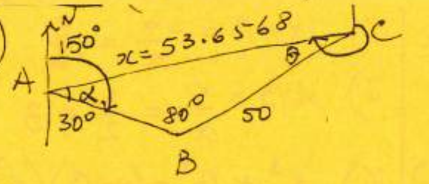
i) $\angle SAB = 30^\circ$ alt. $\angle$'s
$\angle ABN' = \underline{30^\circ}$ on parallel lines
$\therefore \angle ABC = 30^\circ + 50^\circ = 80^\circ$

ii) $AC = x$

$x^2 = 30^2 + 50^2 - 2 \times 30 \times 50 \cos 80^\circ$

$x = 53.6568$ km

iii)



$\dfrac{\sin\alpha}{50} = \dfrac{\sin 80^\circ}{53.6568}$ $\therefore \alpha = 66^\circ 35'$

$\therefore \angle NAC = 150^\circ - \alpha = 83^\circ 25'$

$\therefore \angle NCA = 96^\circ 35'$ (co-int. $\angle$'s on ∥)

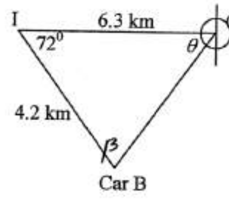
$\therefore$ $\boxed{\text{bearing}}$ is $360^\circ - 96^\circ 35'$
$= 263^\circ 25'$

Q8 Two straight roads intersect at a point I. Car A is 6.3 km along one road and is due east of the intersection. Car B is 4.2 km along the other road on a bearing of 162° from the intersection.

i) Draw a diagram and mark the relevant information on it. **1**

ii) Find the distance between the two cars, correct to 2 significant figures. **1**

iii) Hence or otherwise, find the bearing of Car B from Car A to the nearest degree? **2**

i) 	1	diagram with all given information
ii) $c^2 = a^2 + b^2 - 2bc\cos A$ $AB^2 = 6.3^2 + 4.2^2 - 2 \times 6.3 \times 4.2 \cos 72^\circ$ $AB = 6.4$ km	1	
iii) $\dfrac{\sin A}{a} = \dfrac{\sin B}{b}$ $\dfrac{\sin\theta}{4.2} = \dfrac{\sin 72^\circ}{6.4}$ $\theta = 38^\circ 37'$ or $\beta = 69^\circ$ $\theta = 39^\circ$	2	1 for finding angle
$\therefore 270^\circ - 39^\circ = 231^\circ$ $\therefore$ Bearing of Car B from Car A is 231°T		1 for bearing

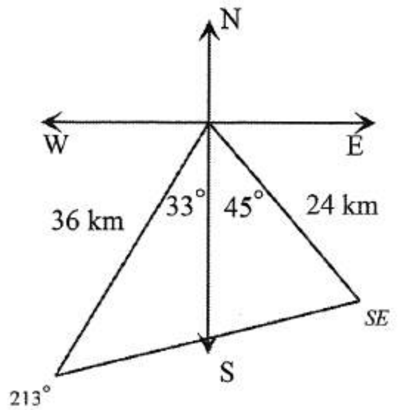
Q9

Two ships leave a port P at the same time. Ship A travels at 12 km/h on a bearing of 213°. Ship B travels in a south easterly direction at a speed of 8 km/h.

i. Draw a diagram to illustrate the relative positions of the two ships after 3 hours, correct to the nearest km. 1

ii. Calculate the distance between the ships after 3 hours, correct to the nearest km. 2

i



ii. $a^2 = b^2 + c^2 - 2bc\cos A$

$d^2 = 36^2 + 24^2 - 2(36)(24)\cos 78$

$d^2 = 1512 \cdot 728598$

$d = 38{\cdot}89381182$

$d = 38{\cdot}9$ nautical miles (3sf)

Q10

Vanessa sets sail on her yacht from Port Pourri (P) on a bearing of 137°. She sails for 15 nautical miles before a storm forces her to change direction and sail on a bearing of 078° to Craggy Cove (C). The diagram shows her course and the point S indicates where she changed direction due to the storm.

The distance PC from Port Pourri to Craggy Cove is 17.4 nautical miles.

(i) Show that $\angle PSC = 121°$. **1**

(ii) Find $\angle PCS$ correct to the nearest degree. **2**

(iii) Hence find the bearing of Port Pourri (P) from Craggy Cove (C). **2**

b)(i) $\angle PSC = (180-137)+78$

$= 43+78$

$= 121°$

b)(ii) Let $\angle PCS = \theta$

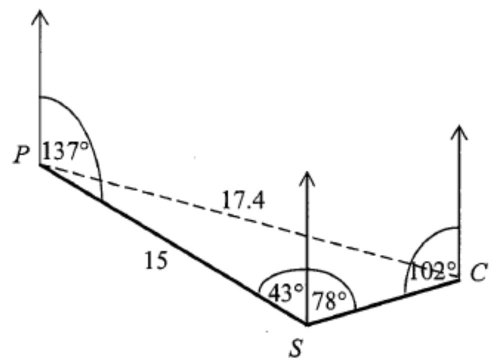
$\frac{\sin\theta}{15} = \frac{\sin 121}{17.4}$

$\sin\theta = \frac{\sin 121}{17.4}\times 15$

$\theta = \sin^{-1}\left(\frac{\sin 121}{17.4}\times 15\right)$

$\theta = 47.6409...$

$\theta = 48°$



b)(iii) Bearing $= 360-(102-48)$

$= 306°T$

Q11 A seaplane flies from a Port (P) to a Quay (Q), 600 kilometres away, on a bearing of 050°T. It then flies on a course of 120°T to a Rock (R), a distance of 850 kilometres from the Quay.

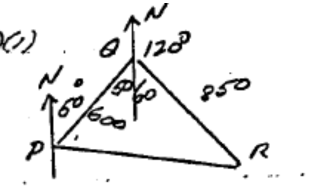
(i) Draw a diagram on your worksheet showing this information. **1**

(ii) Show that $\angle PQR$ is 110°. **1**

(iii) Calculate the shortest distance from the Port to the Rock. (to the nearest kilometre). **1**

(iv) Find the bearing of the Rock from the Port (to the nearest degree). **1**

(c)(i)



(iii) $PR^2 = 600^2 + 850^2 - 2 \cdot 600 \cdot 850 \cdot \cos 110°$

$PR = \underline{1196 \text{ km}}$

(iv) $\cos Q\hat{P}R = \dfrac{600^2 + 1196^2 - 850^2}{2 \times 650 \times 1196}$

$Q\hat{P}R = 42°$

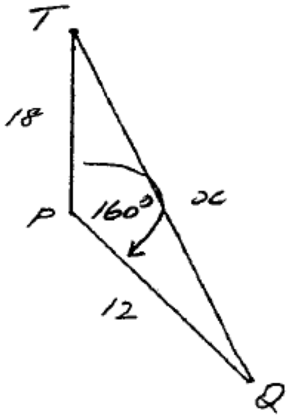
Bearing $\underline{092°T}$

Q12 The course for a yacht race starts from Twofold Bay (Point T). The first leg is due South for a distance of 18 kilometres to Point P. The second leg is on a bearing of 160° for a distance of 12 kilometres to point Q. The third and final leg is in a straight line from Q to T.

(i) Draw a neat sketch showing this information. **1**

(ii) Calculate length of the third leg correct to the nearest one-tenth of a kilometre. **2**

(c) (i)



(ii)

$x^2 = 18^2 + 12^2 - 2 \times 18 \times 12 \times \cos 160°$

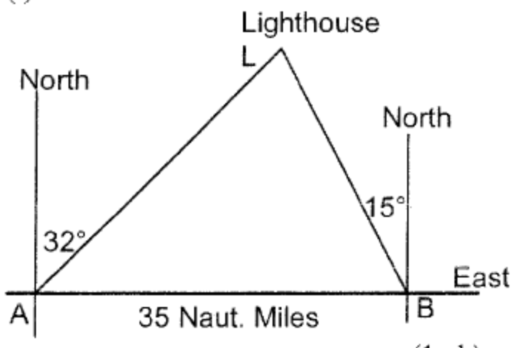
$\underline{x \doteqdot 29.6 \text{ km}}$

Q13 A ship is sailing due East. The navigator took a bearing on a lighthouse and found it to be N 32° E. After sailing a further 35 nautical miles in the same direction the bearing of the lighthouse was N 15° W.

(i) Draw a diagram showing this information. **1**

(ii) Calculate the distance from the ship to the lighthouse when the second bearing was taken. **3**

(i)



(1mk)

(ii)

$\angle LAB = 90° - 32° = 58°$

$(North \perp East)$ (1mk)

$Sim. \angle LBA = 75°$

$\angle ACB = 180° - 133° = 47°$

$(\angle sum\ of\ \Delta LAB)$

$\dfrac{LB}{\sin 58°} = \dfrac{35}{\sin 47°}$ (1mk)

$LB = 40.58nm$ (1mk)

Q14 A ship sails from port A, 30km due East to port B. It then sails a further 18km in the direction 155°T to port C.

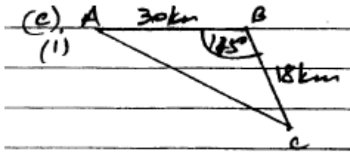
(i) Draw a neat sketch of the above information. **1**

(ii) Show that $\angle ABC = 115°$. **1**

(iii) Find the distance from port A to port C, correct to the nearest metre. **2**

(iv) Find the bearing of port C from port A, correct to the nearest degree. **3**

(c) (i)



(ii) $\angle ABC = 270° - 155°$ ✓

$= 115°$

(iii) $AC^2 = 30^2 + 18^2 - 2.30.18\cos 115°$ ✓

$= 1680.42\ldots$

$AC = 50m.$ ✓

(iv) $\dfrac{\sin A}{18} = \dfrac{\sin 115°}{50}$ ✓

$\sin A = \dfrac{18 \sin 115°}{50}$ ✓

$= 0.326\ldots$

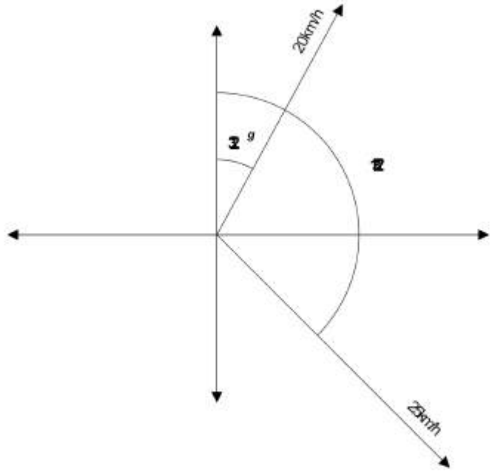
$A = 19°.$ ✓

143

Q15 Zoe and Kobi set out on a bike ride from point P at the same time. One travels at 20km/h along a straight road in the direction 032°T. The other travels at 25km/h along another straight road in the direction 132°T.

ii) Draw a diagram to represent this information. **1**

iii) Find the distance Zoe and Kobi are apart to the nearest kilometre after 3 hours. **2**



let the distance be d

$$d^2 = (20\times 3)^2 + (25\times 3)^2 - 2(20\times 3)(25\times 3)\cos(132-32)$$

$$d = \sqrt{60^2 + 75^2 - (2\times 60\times 75\times \cos(100)}$$

$$\therefore\ d = 104\ km\ (nearest\ km)$$

Q16 The diagram below shows a lighthouse at Q and 40 km from P.
The bearing of the lighthouse, at Q, from P is 115°T.
R is a headland, 30 km from Q and on a bearing of 220°T from Q.

(i) Find the size of $\angle PQR$, correct to the nearest degree. **1**

(ii) Find the distance from P to R, correct to 2 decimal places. **2**

(iii) Find the bearing of R from P. **2**

(i) $\angle PQR = 25° + 50° = 75°$

(ii) cosine Rule

$x^2 = 30^2 + 40^2 - 2\times 30\times 40\times \cos 75°$

$\therefore\ x = 43.35$ km (to 2 dp)

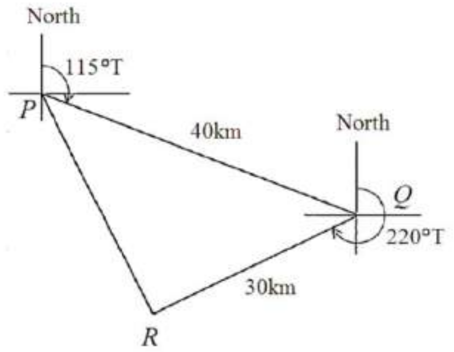
(iii) Let $\angle RPQ = \theta$

$$\frac{\sin\theta}{30} = \frac{\sin 75}{43.35}$$

$$\sin\theta = \frac{\sin 75° \times 30}{43.35}$$

$\theta = 42°$ (to the nearest degrees)

$\therefore$ bearing from R from P is 157°T



Q17 A section of rainforest is to be scoured in the search for new species.
The bearing of landmark A from landmark O is 248°T and is 24 km in distance.
The distance from landmark A to B is 40 km and from landmark B to O is 35 km.

(i) Find the size of $\angle AOB$. **2**

(ii) Hence or otherwise calculate the area of this section of the rainforest. **1**

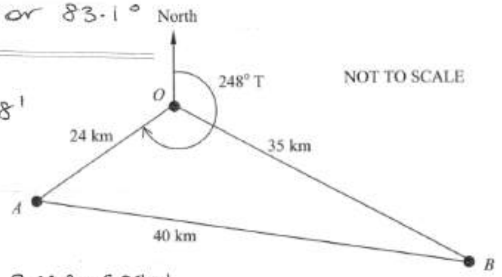
(iii) What is the bearing of landmark O from landmark B? **2**

i) use cosine rule: $\cos A = \dfrac{b^2+c^2-a^2}{2bc}$

$\therefore \cos\angle AOB = \dfrac{24^2+35^2-40^2}{2\times24\times35}$

$= \dfrac{201}{1680}$

$\therefore \angle AOB = 83°7'$ or $83.1°$

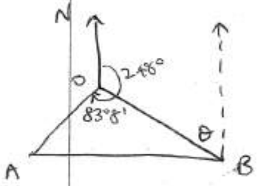


ii) $A = \frac{1}{2}ab\sin C$

$= \frac{1}{2}\times24\times35\times\sin 83°8'$

$= 417\text{m}^2$ (nearest m^2)

iii)



$\angle NOB = 248° - 83°8'$

$= 164°52'$

$\therefore \theta = 180 - 164°52'$ (co-int ∠s)

$= 15°8'$

$\therefore$ Bearing of O from B $= 360° - 15°8'$

$= \underline{\underline{344°52'\text{T}}}$.

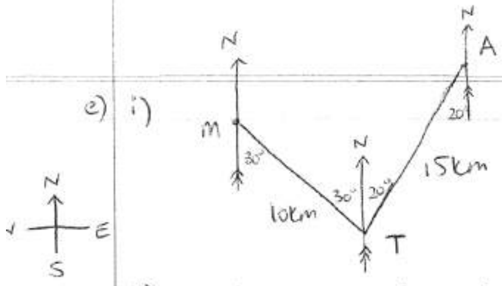
Q18 A boat sails 10 km from a Marina on a bearing of S30°E. It then sails 15 km on a bearing of N20°E before dropping anchor.

i) Draw a diagram showing all key information about the boat's journey. 1

ii) How far is the boat from the Marina? Give your answer correct to the nearest metre. 2

iii) What is the bearing of the Marina from the boat where it is anchored? 2

e) i)



M = Marina
T = turning pt
A = anchored pt.

ii) $AM^2 = 10^2 + 15^2 - 2(10)(15)\cos 50°$

$= 132.1637\ldots$

$AM = 11.496$ km (nearest m)

iii) bearing °T $= 180° + 20° + \angle MAT$

$\dfrac{\sin\angle MAT}{10} = \dfrac{\sin 50°}{AM}$

$\angle MAT = 41.79°$ (2dp)

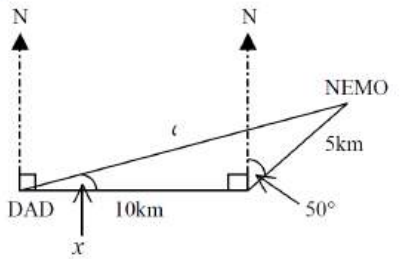
$\therefore$ bearing of Marina from boat is 242°T (nearest deg)

Q19 Nemo swims 10km due East from his Dad and then 5km on a bearing of 050°T.

i) Find the distance from Nemo to his Dad to the nearest km.

ii) Find the bearing of Nemo from his Dad to the nearest degree. **4**

c)



i) Using cosine rule:

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\therefore d^2 = 10^2 + 5^2 - 2\times10\times5\times\cos140°$$

$$= 125 + 76\cdot604444...$$

$$= 201\cdot604444...$$

$$d = 14\cdot198747.....$$

$= 14$km (nearest km)

ii) Using sine rule:

$$\frac{\sin x}{5} = \frac{\sin 140°}{14\cdot198747...}$$

$$\sin x = \frac{5\sin140°}{14\cdot198747...}$$

$$= 0\cdot226353...$$

$$x = 13\cdot082488...$$

$= 13°$ (nearest deg)

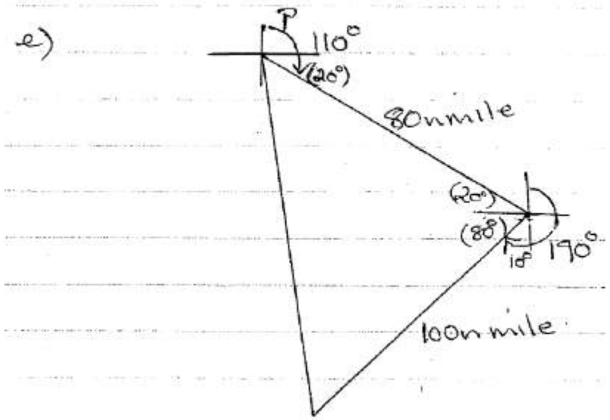
Bearing of Nemo from Dad is
(90 _ 13)°T = 077°T.

Q20 A boat leaves port P and steams 80 nautical miles on a bearing of 110°.
It then turns to a bearing of 190° and steams a further 100 nautical miles.

i) Draw a diagram illustrating this information. **1**

ii) How far is the boat from P to the nearest nautical mile? **2**

e)



$d^2 = 80^2 + 100^2 - 2(80)(100)\cos 100°$

$d \doteqdot 138\cdot4859...$

the distance is approx

$\therefore$ 138 nautical miles.

Q21

Keiran and Marcus set out on a bike ride from point P at the same time. One travels at 25km/h along a straight road in the direction 042° T. The other travels at 30km/h along another straight road in the direction 142° T.

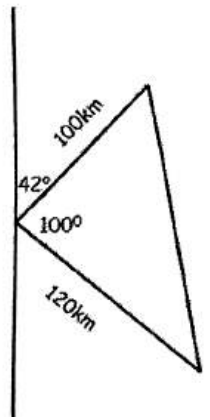
(i) Draw a diagram to represent this information. **1**

(ii) Find the distance Keiran and Marcus are apart to the nearest kilometre after 4 hours. **2**

12 (b) (i) (1 mark)
Outcomes Assessed: P4
Targeted Performance Bands: 2-3

Criteria
• One mark for correct diagram

Answer



12 (b) (ii) (2 marks)
Answer

$d^2 = 100^2 + 120^2 - 2\times 100\times 120\times Cos100^\circ$

$d^2 = 28567.55626$

$d = 169.0193961$

$d = 169km$

Q22

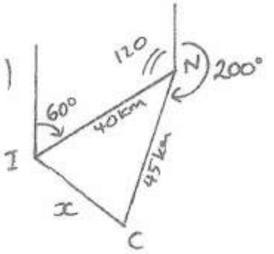
A boat leaves the Island of Ischia and travels 40 km on a bearing of $060°T$ to Naples. It then turns to travel a further 45 km to the Island of Capri on a bearing of $200°T$.

(i) Find the size of $\angle INC$. **1**

(ii) Find the distance between the islands of Ischia and Capri to the nearest kilometre. **2**

(iii) Find the area of ΔINC. Give your answer in square kilometres to 2 significant figures. **2**

(c) (i)



$\angle INC = 360 - 200 - 120$

$= 40°$ ✓

(ii) $x^2 = 40^2 + 45^2 - 2\times 40\times 45\times \cos 40$ ✓

$= 867.24$

$x = \sqrt{867.24} \doteqdot 29\text{ km}$ ✓

(iii) Area $= \frac{1}{2}\times 40\times 45\times \sin 40°$ ✓

$\doteqdot 578.5088\ldots$

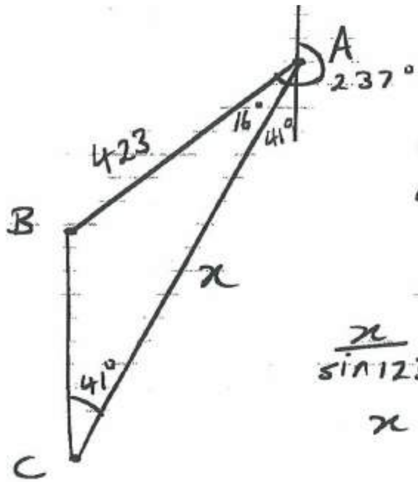
$\doteqdot 580\text{ km}^2$ (2 sig fig) ✓

Q23 A ship sails from point A on a bearing of 237° for a distance of 423 kilometres to point B. The ship then turns and sails due south to point C. The bearing of A from C is 041°.

(i) Draw a diagram showing this information **1**

(ii) Find the size of $\angle BAC$, to the nearest degree **1**

(iii) Calculate the distance (to the nearest km) the ship must travel back to point A from C. **2**



(i) ✓ $\boxed{\text{Com } 1}$

(ii)
$\angle BAC = 237 - 41 - 180$
$= 16°$ ✓

(iii) $\angle ABC = 123°$

$$\frac{x}{\sin 123°} = \frac{423}{\sin 41°}$$ ✓

$x = 541$ km ✓

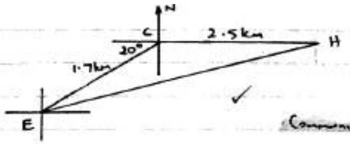
Q24 Holly and Ellen go bushwalking. They leave their campsite and set out on different directions. Holly walks due East for 2.5km while Ellen walks 1.7km on a bearing of 250°T

(i) Draw a diagram showing the above information. **1**

(ii) Calculate the distance between Holly and Ellen. **2**

(iii) Calculate the bearing of Holly from Ellen. **3**

b) i)



✓ Communication 1

ii) $\angle ECH = 250° - 90°$
$= 160°$

$\therefore EH^2 = 2.5^2 + 1.7^2 - 2 \times 2.5 \times 1.7 \times \cos 160°$ ✓
$= 17.127\ldots$

$EH = \sqrt{17.127\ldots}$
$= 4.1$ (to 1 dec. pl.) ✓

$\therefore$ the distance between Holly and Ellen is 4.1km.

iii) Let $\angle CEH = \theta$

$$\frac{\sin\theta}{2.5} = \frac{\sin 160}{4.1}$$ ✓

$$\sin\theta = \frac{2.5 \times \sin 160}{4.1}$$

$= 0.2066\ldots$

$\therefore \theta = \sin^{-1}(0.2066\ldots)$
$= 12°$ (to nearest degree) ✓

$\therefore$ bearing $= 70 + 12$
$= 82°$

$\therefore$ the bearing of Holly from Ellen is 082°T ✓

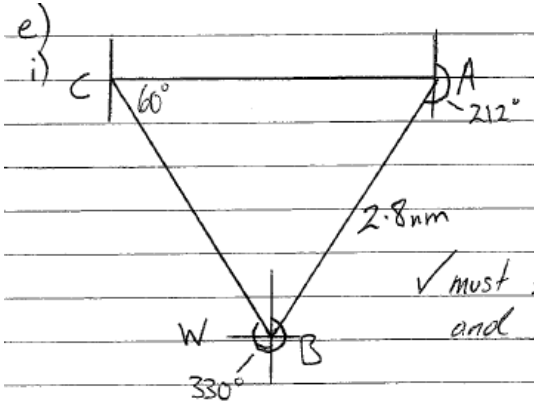
Reasoning – 3

Q25 A yacht sails from a port A on a course bearing of $212°$ for 2.8 nautical miles to B. It then turns and sails on a course bearing of $330°$ to a point C, due west of A.

(i) Find $\angle ACB$. **1**

(ii) Hence, find to the nearest tenth of a nautical mile, the distance BC. **2**

e) i)



C1

✓ must show b
and length

ii) $\angle CBW = 330° - 270°$

$= 60°$

$\angle ACB = \angle CBW$ (alt. $\angle$s = on ∥ lines)

$= 60°$ ✓

iii) $\angle CAB = 270 - 212$

$= 58°$ ✓

$$\frac{BC}{\sin 58°} = \frac{2.8}{\sin 60°}$$

$BC = 2.7\,nm$ ✓

Q26 A ship sails from port A on a bearing of 050^0 for $230km$ to port B. It then changes its bearing to 120^0 and continues sailing for $400km$ to port C.

i) Show all the given information on your diagram.

ii) Show that $\angle ABC = 110^0$. (1mark)

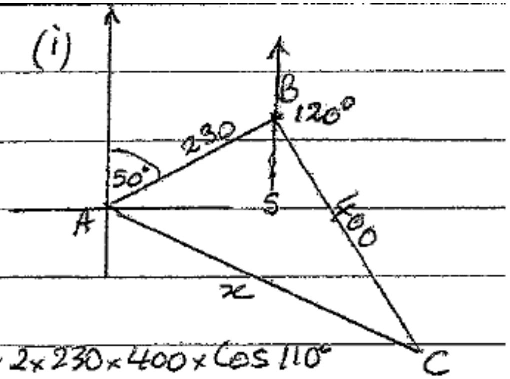
iii) Calculate how far, to the nearest km, the ship is from its starting point. (2marks)

ii) $\angle ABS = 50°$

$\angle CBS = 60°$

$\therefore \angle ABC = 110°$

(i)



(iii) $x^2 = 230^2 + 400^2 - 2 \times 230 \times 400 \times \cos 110°$

$= 212900 - 184000\,(-0.34202)$

≈ 275831.7

$\Rightarrow x \approx 525$ km.